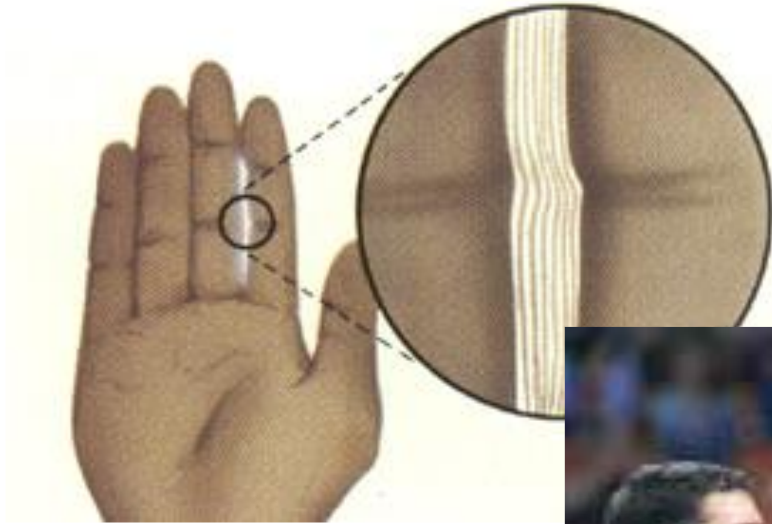
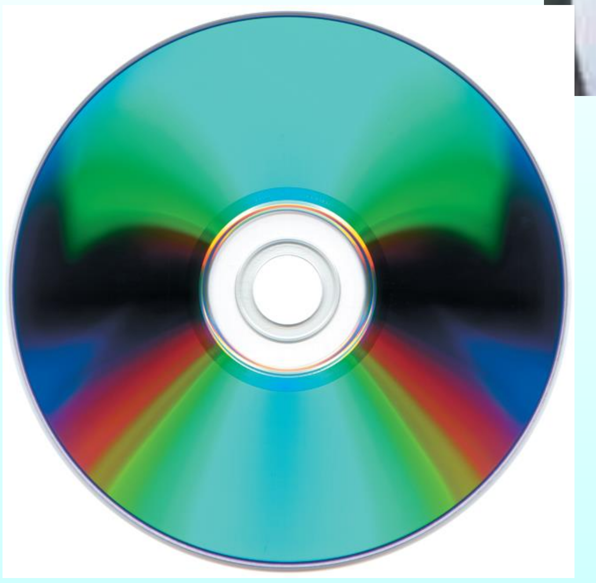
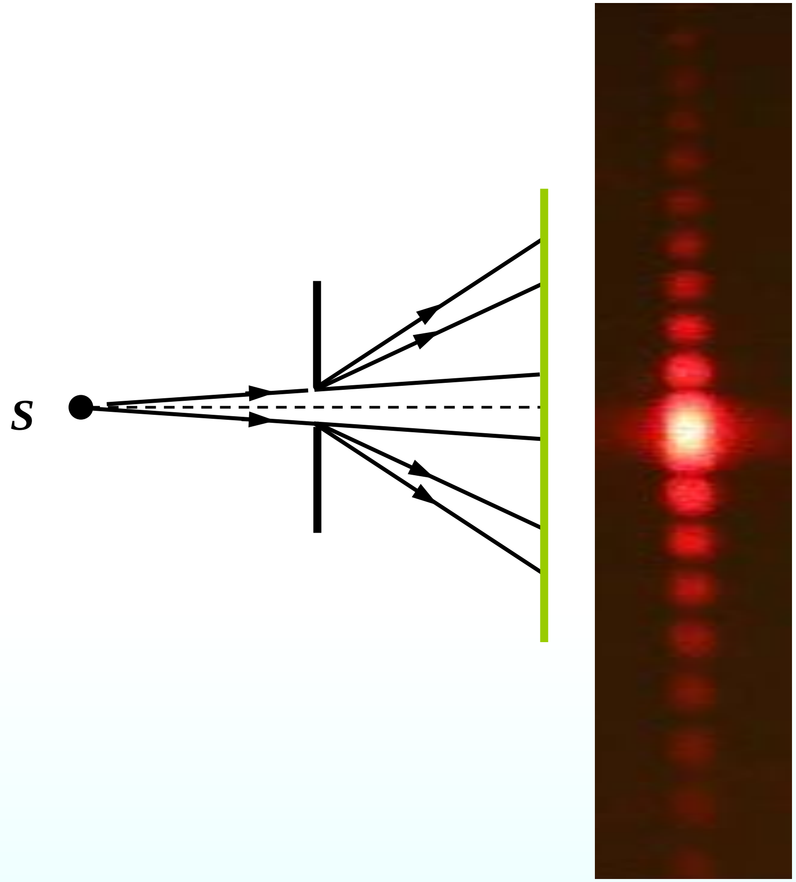
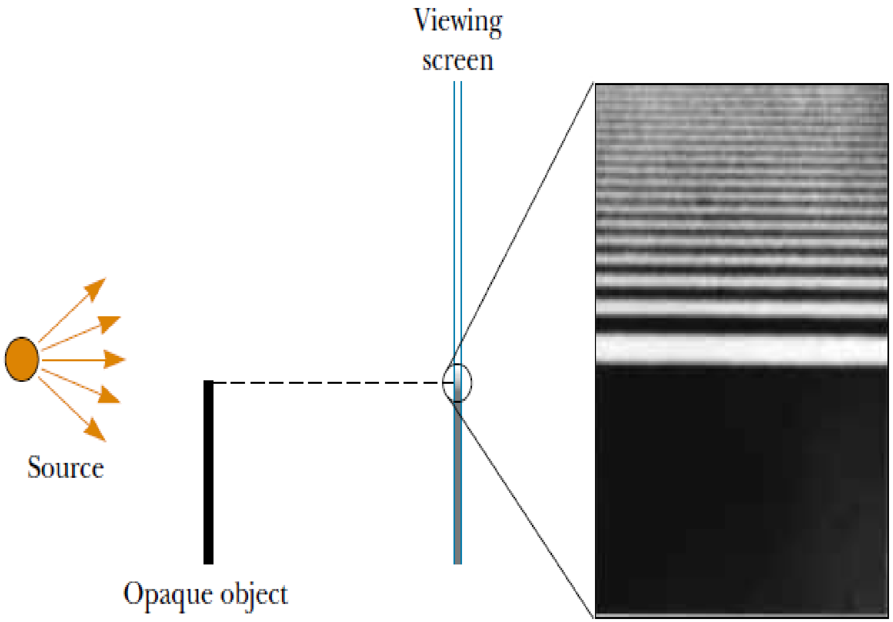
Key words And Terms

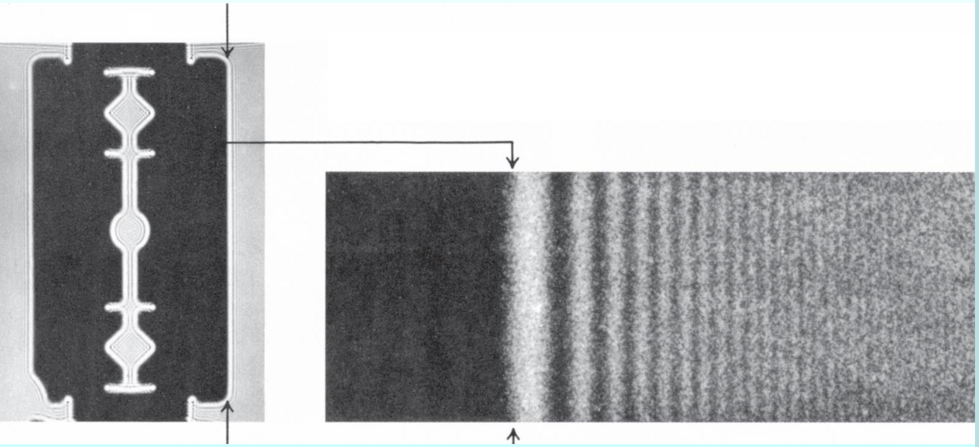
Huygens-Fresnel principle	**惠更斯-菲涅耳原理**
single-slit diffraction	**单缝衍射**
angle of diffraction	**衍射角**
half-wave zone	**半波带**
circular aperture diffraction	**圆孔衍射**
Airy disk	**爱里斑**
angle of minimum resolution	**最小分辨角**
grating	**光栅**
grating constant	**光栅常数**
missing order	**缺级**
X-ray diffraction	**X射线衍射**
Bragg equation	**布拉格方程**



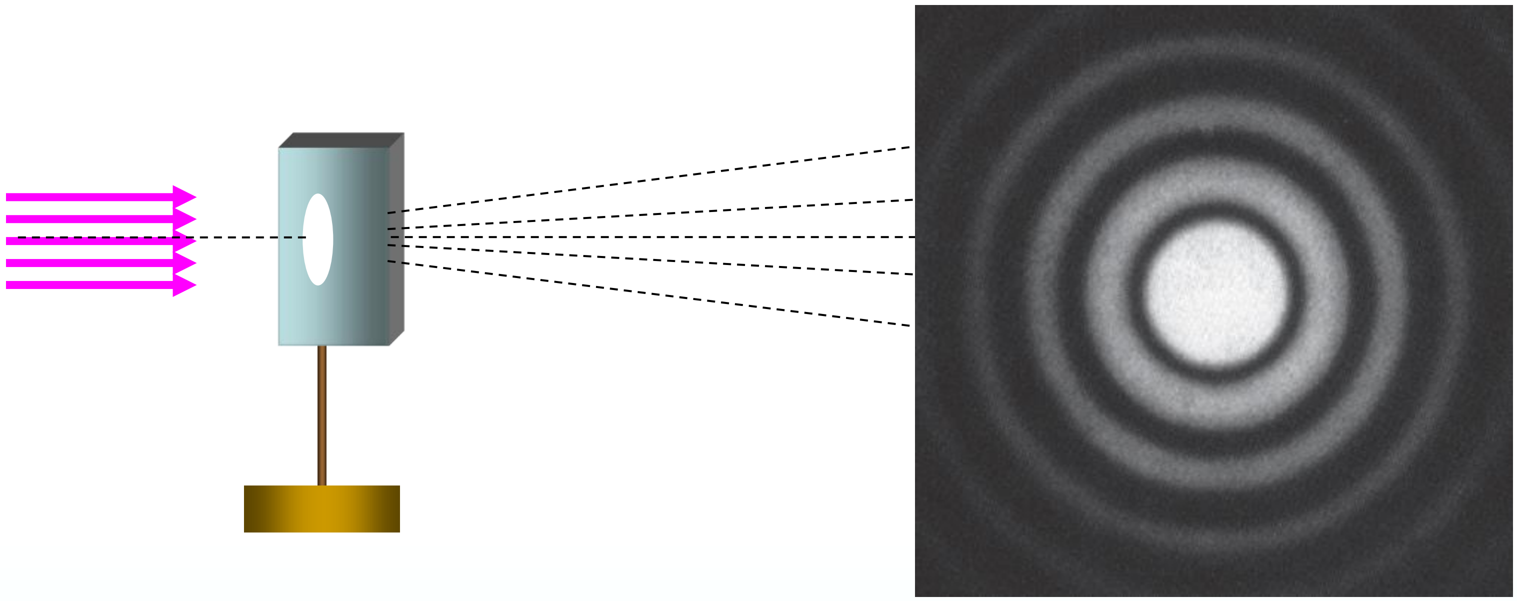


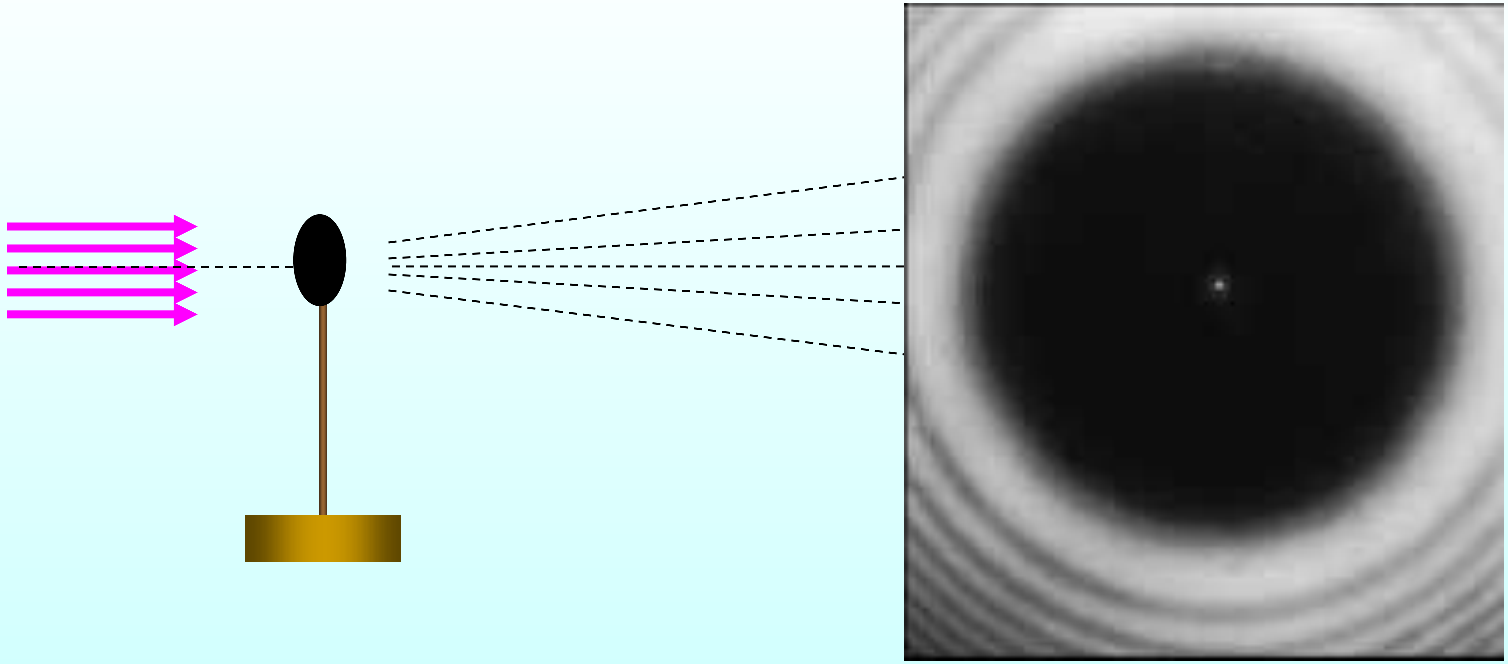






Light from a small source passes by the edge of an opaque object. In reality, light bends around the top edge of the object and enters this region. A diffraction pattern consisting of bright and dark fringes appears in the region above the edge of the object.





Chapter 18 Diffraction

18.1 The phenomenon of diffraction

18.2 Diffraction by a single slit

18.3 Limits of Resolution; Circular Apertures

18.4 Diffraction in the Double-slit Experiment

18.5 Diffraction Grating

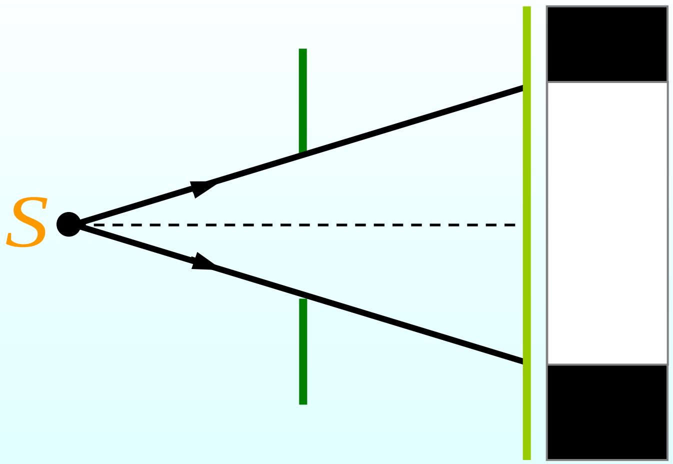
***18.6 *X*-Ray diffraction**

Chapter 19 Diffraction

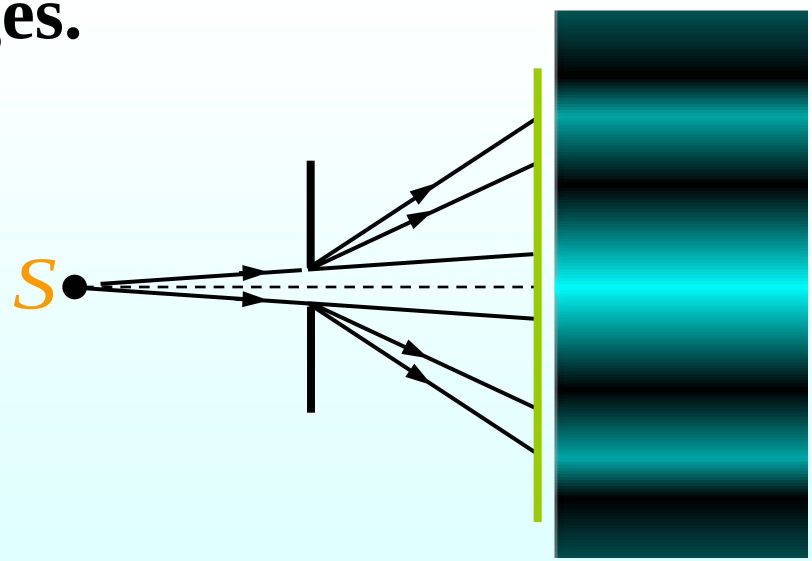
19.1 The phenomenon of diffraction

1. The phenomenon of diffraction

The waves, as they pass the small obstacle, bent around behind it into the "shadow region". In general, diffraction occurs when waves pass through small openings, around obstacles, or past sharp edges.



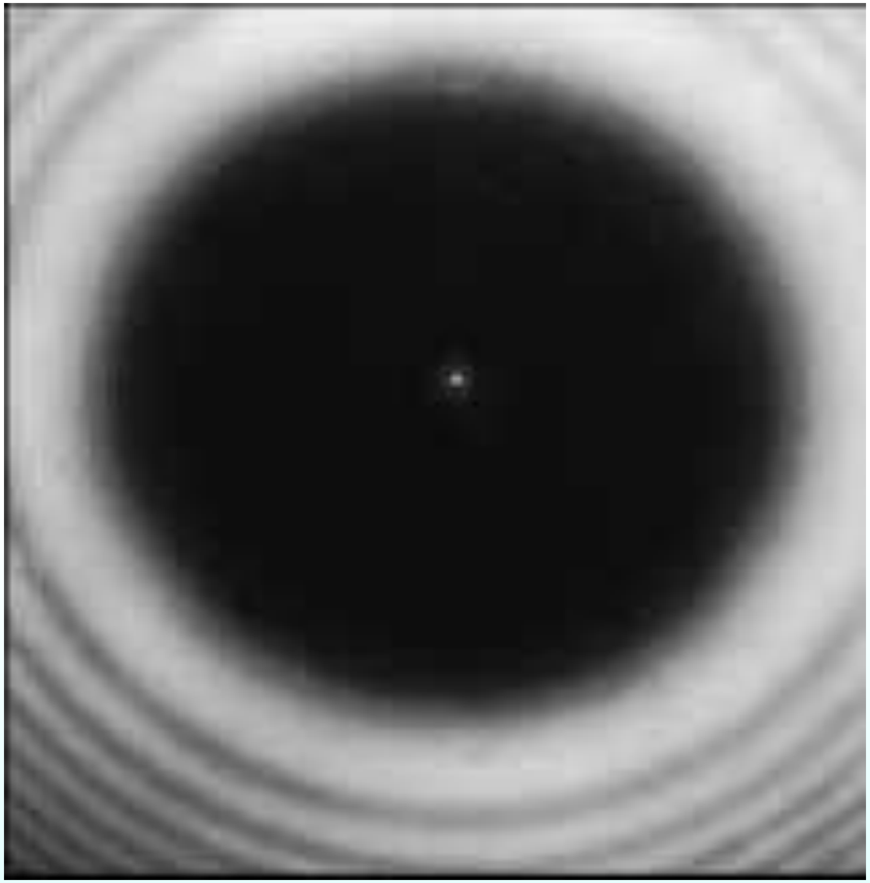
Large obstacle



Small obstacle

An opening of dimensions similar of the wavelength

Fresnel Bright Spot



Diffraction pattern created by the illumination of a penny, with the penny positioned midway between screen and light source.

菲涅尔(1788～1827)
法国物理学家

“物理光学之父”

2. 惠更斯——菲涅耳原理

惠更斯原理

在波的传播过程中，波前上的每一点都可看作是发射子波（次波）的波源，在其后的任一时刻，这些子波的包迹就成为新的波阵面。

菲涅耳假定

波在传播过程中，从同一波阵面上各点发出的子波，经传播而在空间某点相遇时，产生相干叠加。

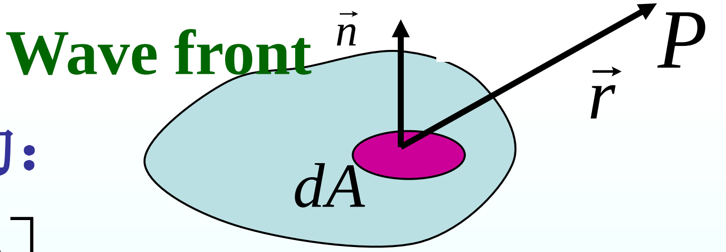
•菲涅耳还指出：

(1) 波面是一个等位面，其上各点相位相同。

(2) 次波在P点的振幅与距离r成反比。

(3) dA面元所发出的次波的振幅与dA面积成正比。
随dA面元的法线与r之间的夹角θ 增大而减小。

•数学表达式



波面S上所有面元dA在P点的合振动：

$$E = C\int \frac{K(\theta)}{r}\cos\left[2\pi\left(\frac{t}{T}-\frac{r}{\lambda}\right)\right]dA$$

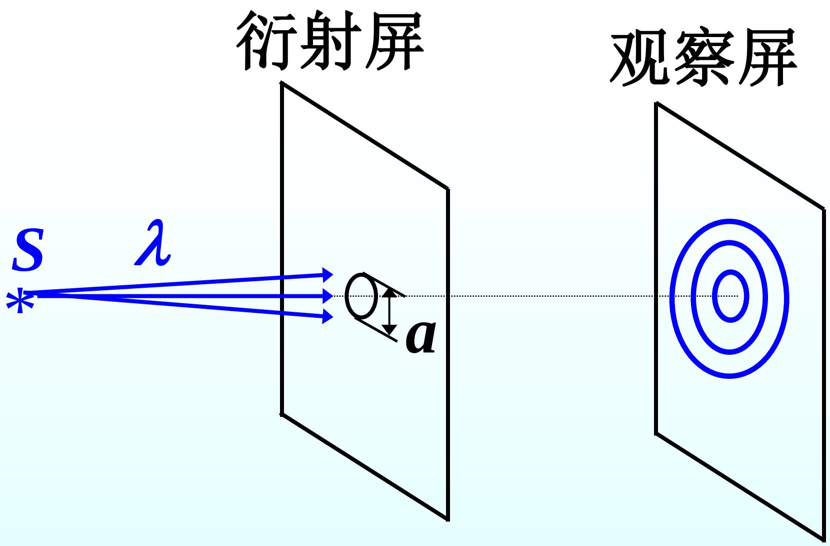
式中C为比例系数，$K(\theta)$为倾斜因数，$\theta \uparrow \rightarrow K(\theta)\downarrow$

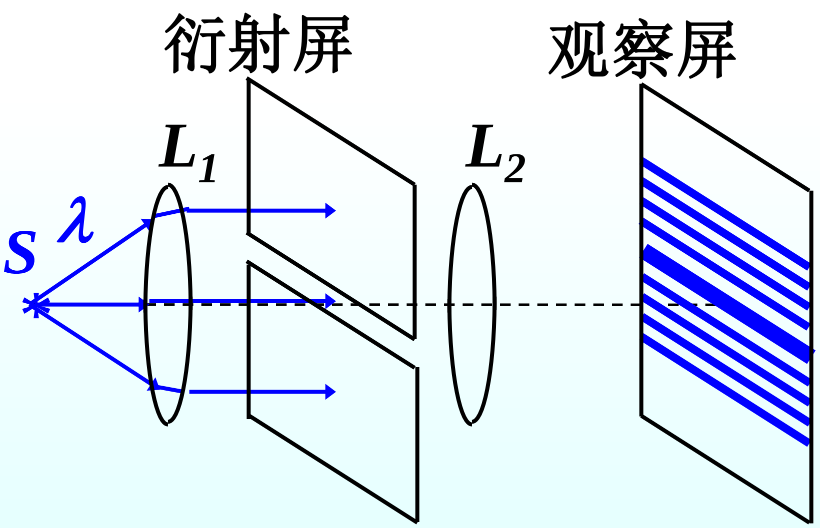
$$K(\theta) = \begin{cases} 0 & \text{当 } \theta \geq \dfrac{\pi}{2} \text{ 时} \\ 1 & \text{当 } \theta = 0 \text{ 时} \end{cases}$$

3. 衍射的分类:

(1) Fresnel diffraction 菲涅耳衍射 ——近场衍射

衍射屏离光源或接收屏为有限远



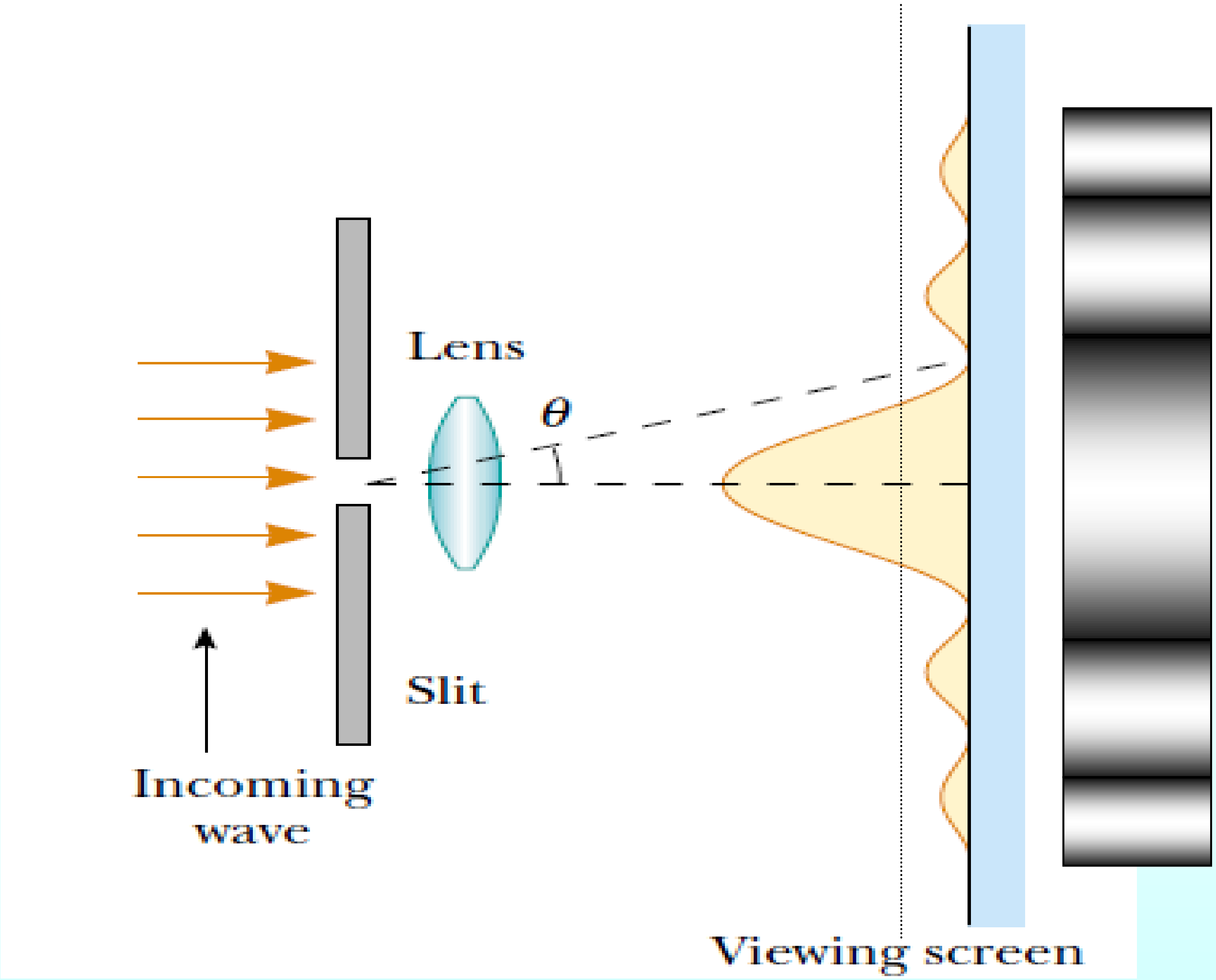


(2)Fraunhofer diffraction 夫琅禾费衍射 ——远场衍射

衍射屏与光源和接收屏为无限远

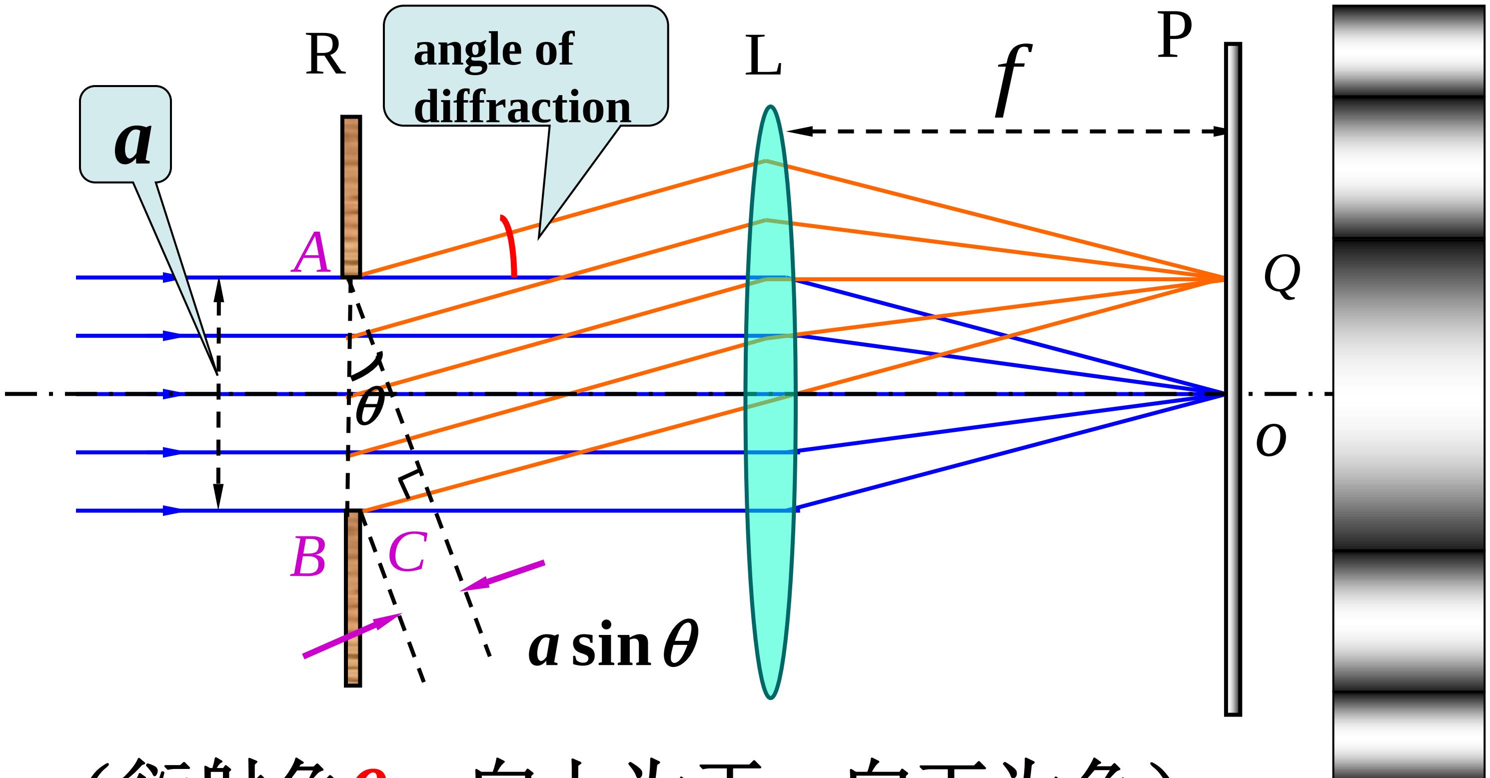
19.2 Diffraction by a single slit

Parallel rays of monochromatic light pass through the narrow slit.



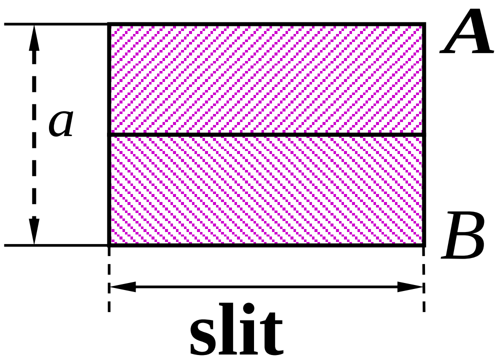
1. Intensity in single-slit diffraction pattern

夫琅禾费单缝衍射

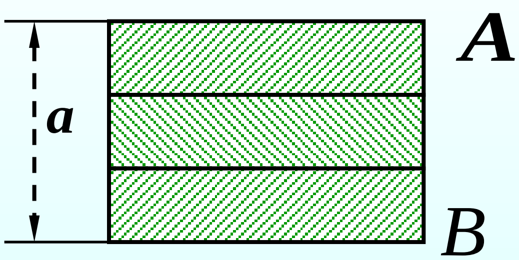


（衍射角***θ***：向上为正，向下为负）

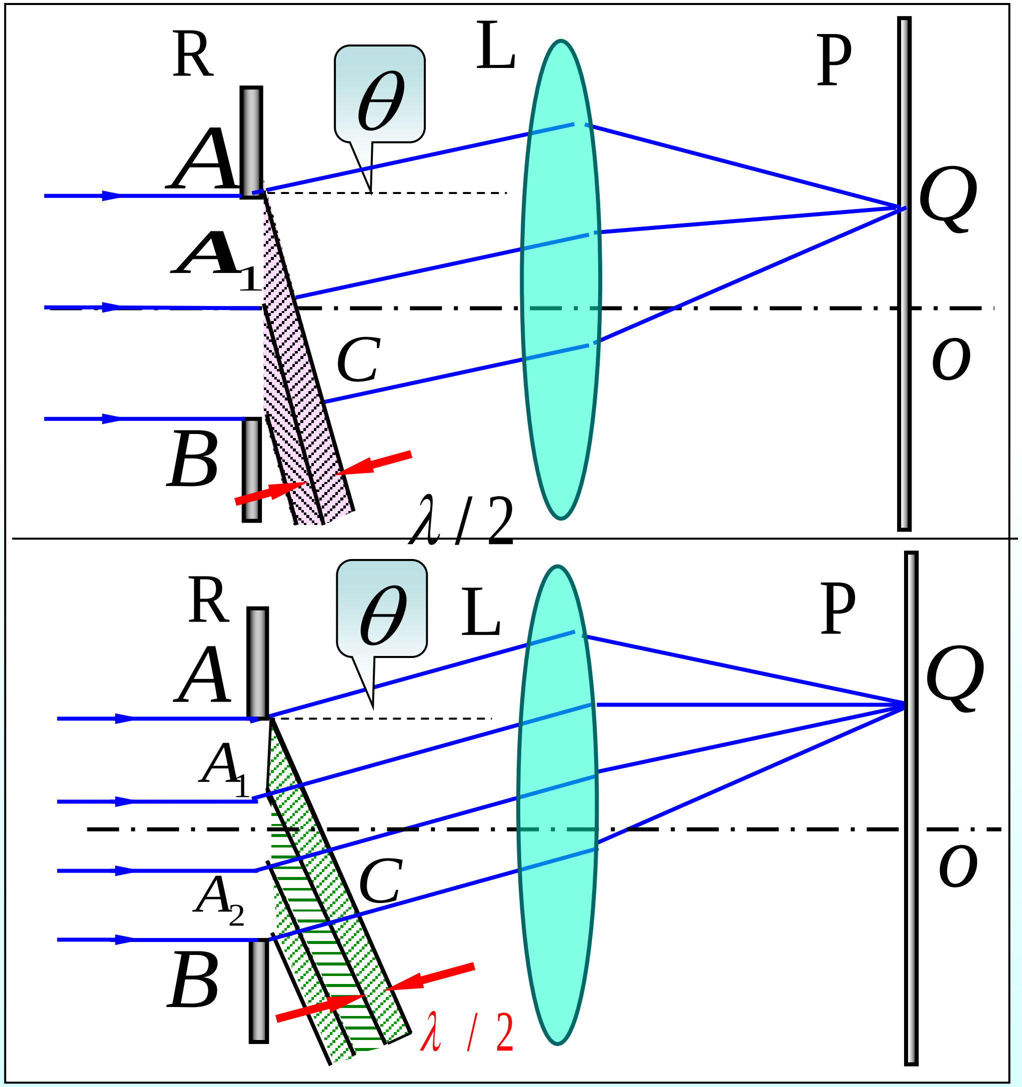
We divided the surface of the slit into several equal parts, and adthe two jacent part rays have path difference λ/2 . The intensity of light rays coming from each parts are equal.



$$a\sin\theta = 2\frac{\lambda}{2}$$

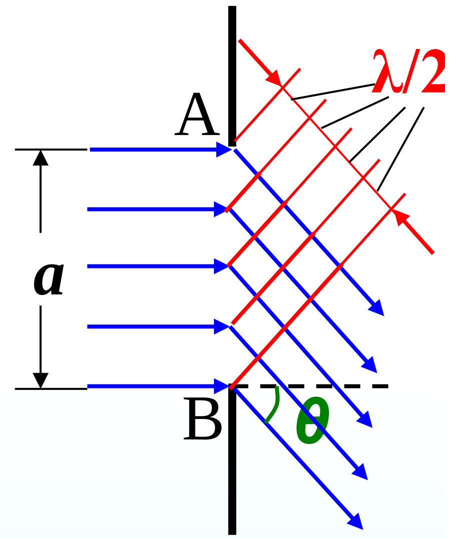


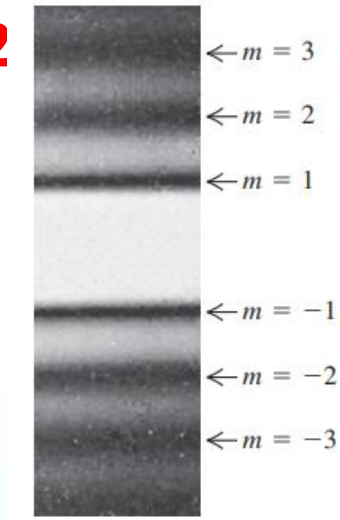
$$a\sin\theta = 3\frac{\lambda}{2}$$



2 .The positions on which all rays travel with angle θ

$$a\sin\theta = N\frac{\lambda}{2}$$





Dark fringe

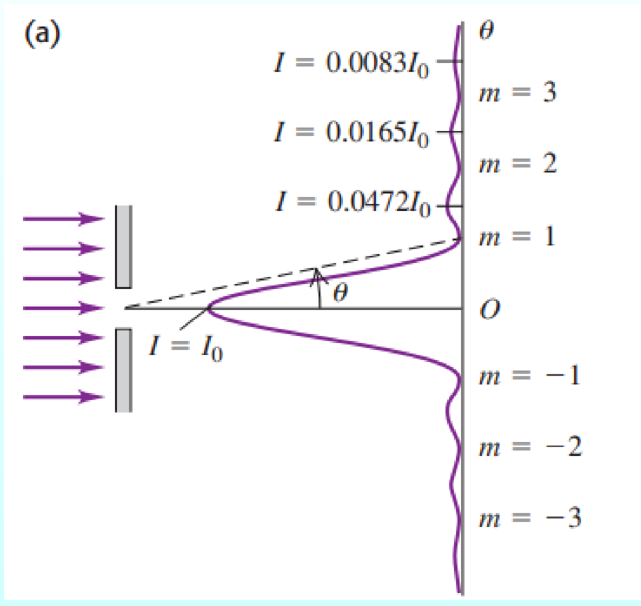
$$a\sin\theta = \pm 2m\frac{\lambda}{2} = \pm m\lambda, \quad m = 1, 2, 3\cdots$$

Bright fringe

$$a\sin\theta = \pm(2m+1)\frac{\lambda}{2}, \quad m = 1, 2, 3\cdots$$

Center bright fringe 中央明纹

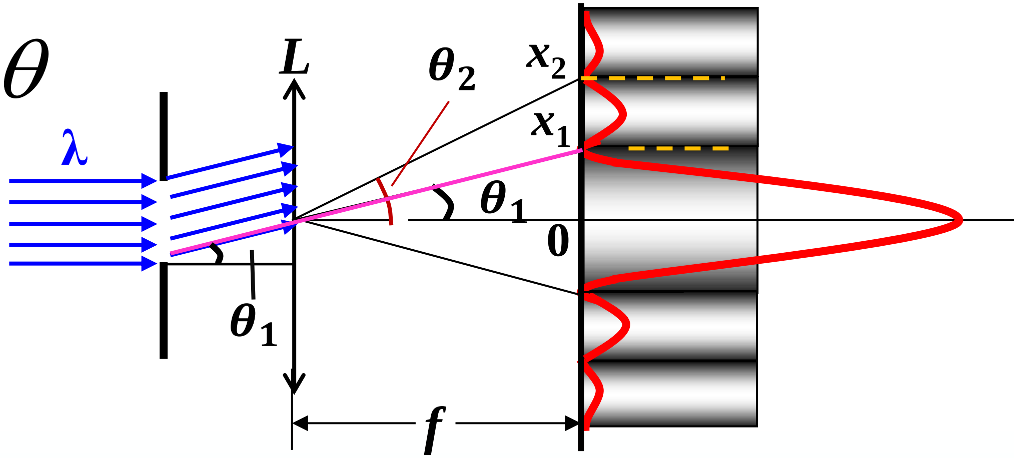
$$a\sin\theta = 0$$



3. The positions of bright and dark bands on the screen

$$a >> \lambda \quad \sin\theta \approx \tan\theta \approx \theta$$

$$x = f\tan\theta$$



$$a\sin\theta = \pm m\lambda, \qquad m = 1, 2, 3\cdots$$

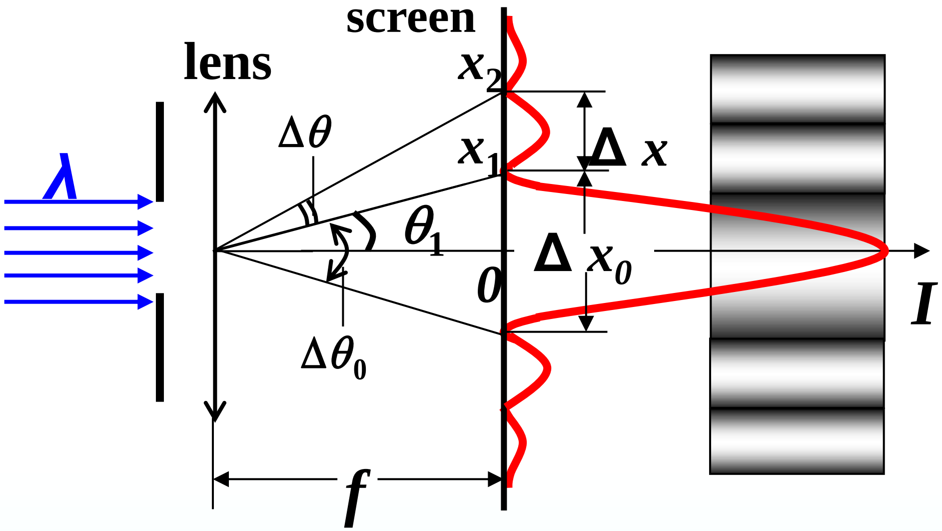
$$a\frac{x}{f} = \pm m\lambda, \qquad x_D = \frac{m\lambda}{a} f$$

$$a\sin\theta = \pm(2m+1)\frac{\lambda}{2}, \qquad m = 1, 2, 3\cdots$$

$$a\frac{x}{f} = \pm(2m+1)\frac{\lambda}{2}, \qquad x_B = (2m+1)\frac{\lambda}{2a} f$$

4. angular-width and line-width

(1)Center bright fringe



Angular-width

$$\Delta\theta_0 = 2\theta_1 \approx 2\frac{\lambda}{a}$$

Line-width $\Delta x_0 = f\Delta\theta_0 = 2f\frac{\lambda}{a} \propto \frac{\lambda}{a}$

(2)Other fringes

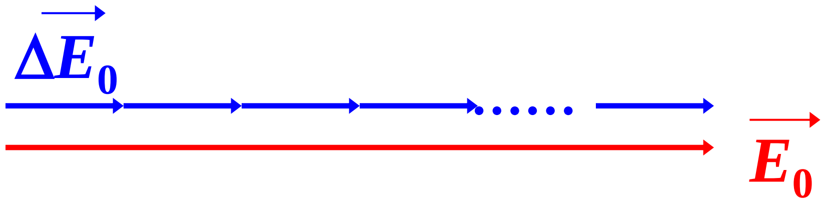
Angular-width $\Delta\theta = \frac{\lambda}{a}$

Line-width $\Delta x = \frac{f\lambda}{a} = \frac{1}{2}\Delta x_0$

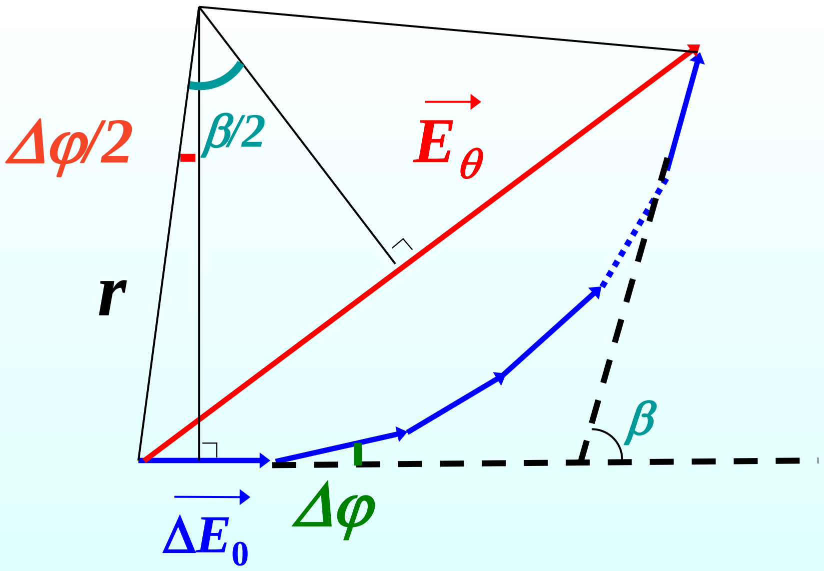
* *Intensity in single-slit diffraction pattern*

At the center $\quad \theta = 0, \ \Delta\varphi = 0$

$$E_0 = N\,\Delta E_0$$



Other points: $\quad E_\theta < E_0$ (*n is large*)



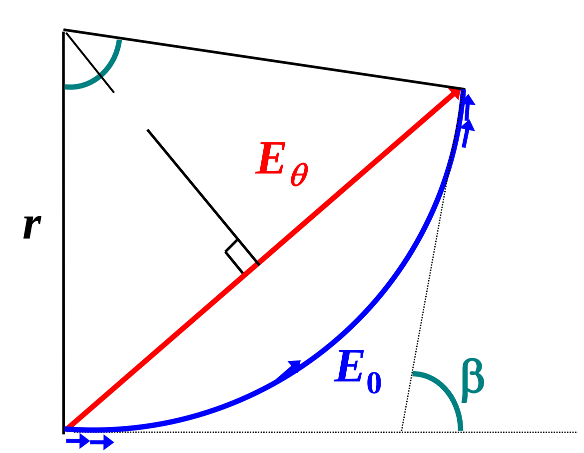
$$E_\theta = 2r\sin\frac{\beta}{2}$$

$$r = \frac{\Delta E_0/2}{\sin(\Delta\varphi/2)}$$

$$\beta = N\Delta\varphi = \frac{2\pi}{\lambda}a\sin\theta$$

$$E_\theta = 2\frac{\Delta E_0/2}{\sin(\Delta\varphi/2)}\sin\frac{\beta}{2} = \Delta E_0\frac{\sin(\beta/2)}{\beta/2N} = E_0\left(\frac{\sin\beta/2}{\beta/2}\right)$$

***Intensity is proportional to the square of the wave amplitude**

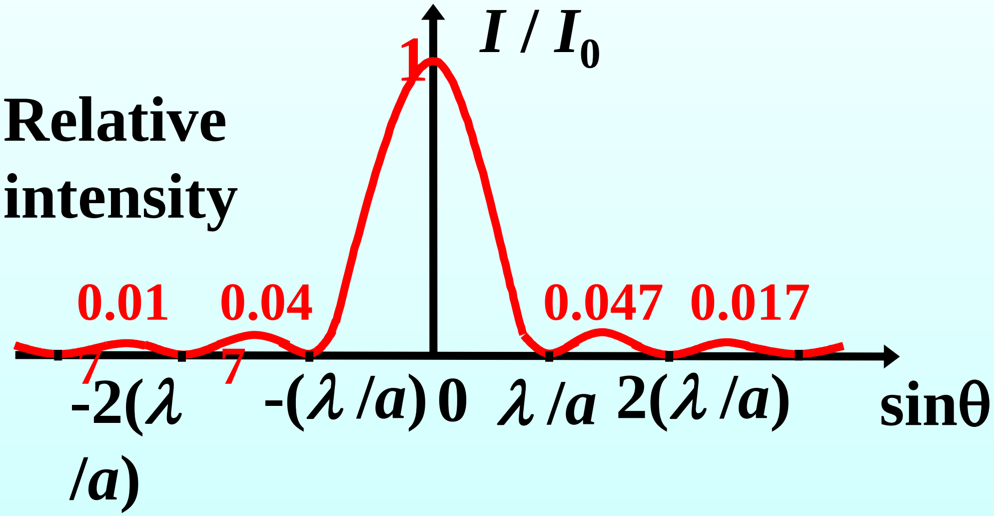


$$I_\theta = I_0\left(\frac{\sin\beta/2}{\beta/2}\right)^2 = I_0\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right]^2$$

(a) For very small angles, $\sin\beta/2 \approx \beta/2$

The maximum at the center $I = I_0$

Relative intensity



$$I_\theta = I_0\left(\frac{\sin\beta/2}{\beta/2}\right)^2 = I_0\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right]^2$$

(b) Dark fringe

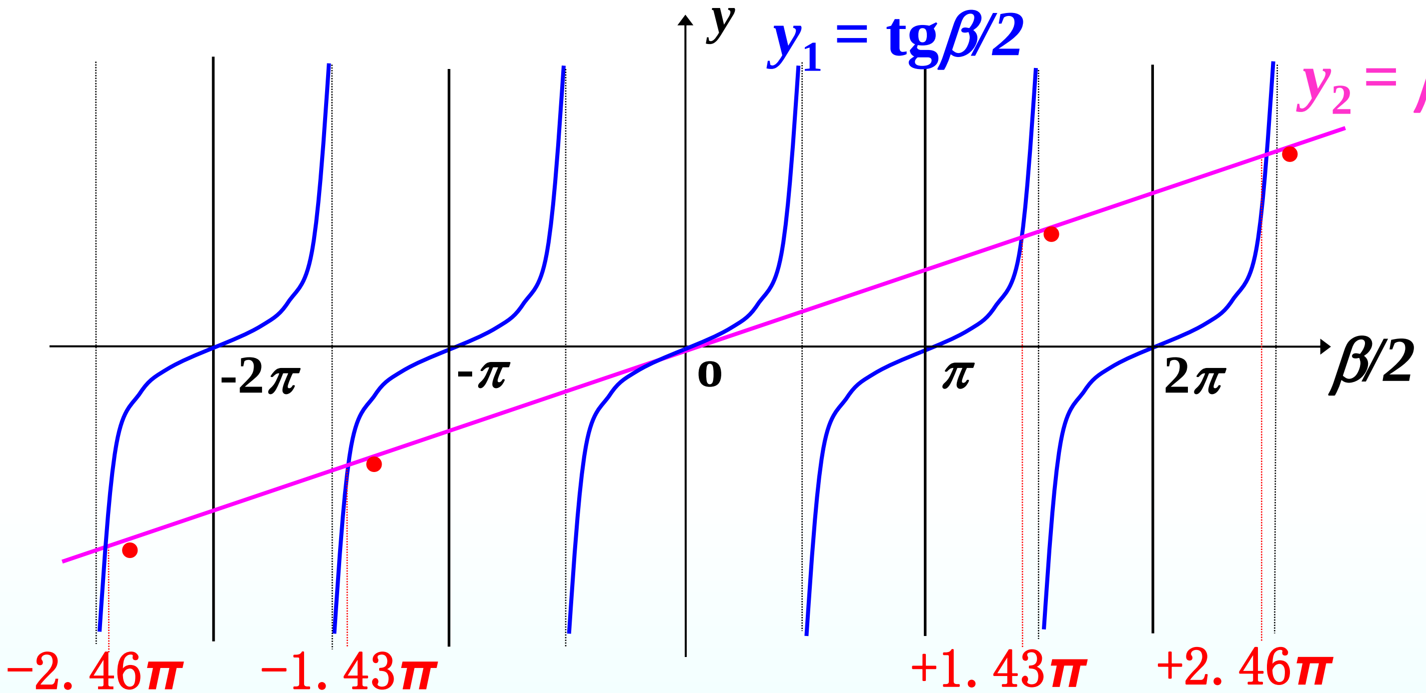
$$I_\theta = 0 \qquad \frac{\sin\beta}{2} = \frac{\pi a\sin\theta}{\lambda} = m\pi$$

$$a\sin\theta = \pm m\lambda, \qquad m = 1, 2, 3\cdots$$

$$N\Delta\varphi = \pm 2k\pi \rightarrow a\sin\theta = \pm k\lambda$$

(c) Bright fringe

$$\frac{dI}{d\beta} = 0 \rightarrow \mathrm{tg}\,\beta/2 = \beta/2$$



$$\beta/2 = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \cdots$$

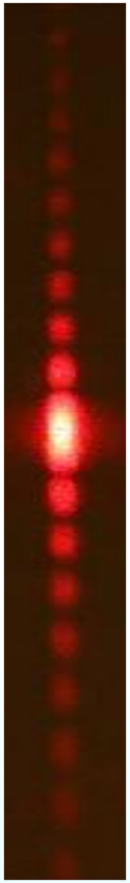
$$\boldsymbol{a \sin\theta = \pm 1.43\lambda, \pm 2.46\lambda, \pm 3.47\lambda, \cdots}$$

Intensity at secondary maxima.

$$I_\theta = I_0\left(\frac{\sin\beta/2}{\beta/2}\right)^2 = I_0\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right]^2$$

$$\frac{\beta}{2} = \frac{\pi a\sin\theta}{\lambda} \approx \left(m+\frac{1}{2}\right)\pi \qquad m = 1, 2, 3\cdots$$

$$I_\theta = \frac{I_0}{\left(m+\frac{1}{2}\right)^2\pi^2} \qquad m = 1, 2, 3\cdots$$

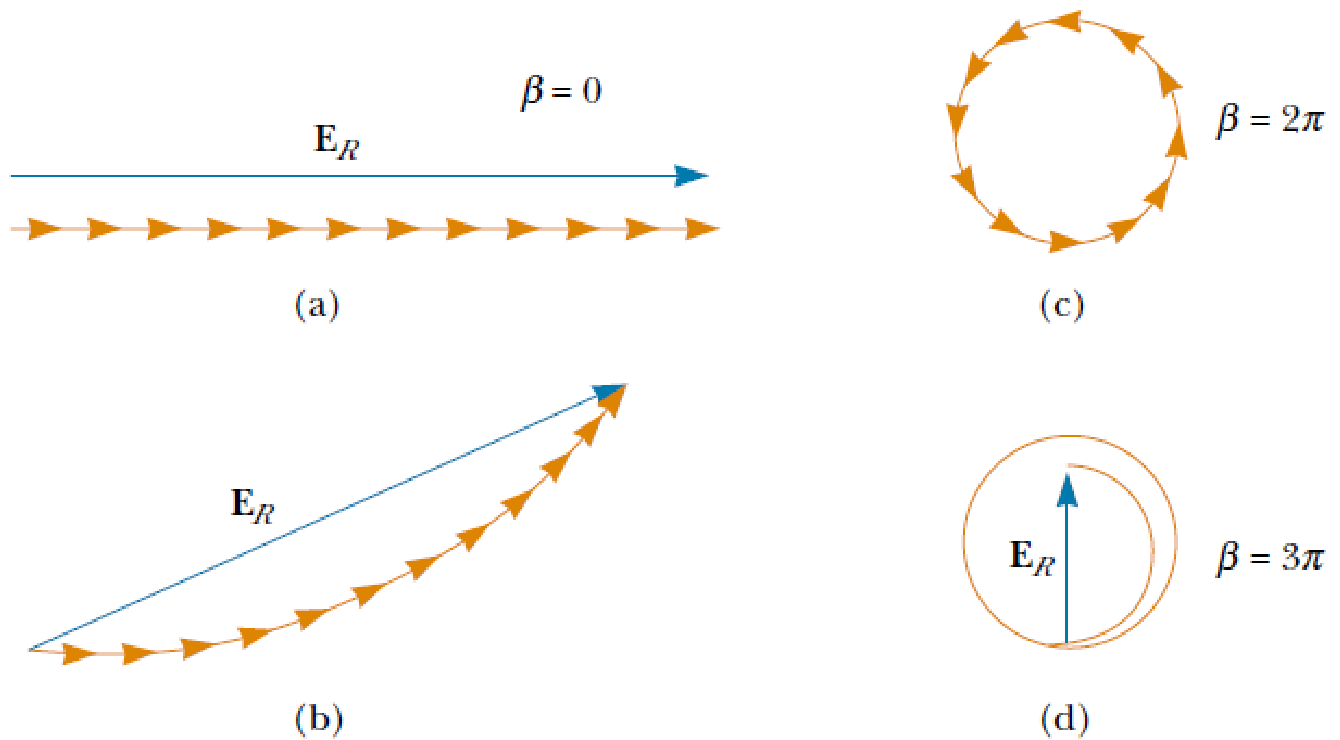


The first two secondary maxima to either side of the center maximum

***m*=1** $$I_\theta = \frac{I_0}{22.2} = 0.045 I_0$$

***m*=2** $$I_\theta = \frac{I_0}{61.7} = 0.016 I_0$$

When $N \to \infty$, the length of the arc is E_0



$$\beta = N\Delta\varphi = \pm 2k\pi \to a\sin\theta = \pm k\lambda$$

Phasor diagrams for obtaining the various maxima and minima of a single-slit diffraction pattern.

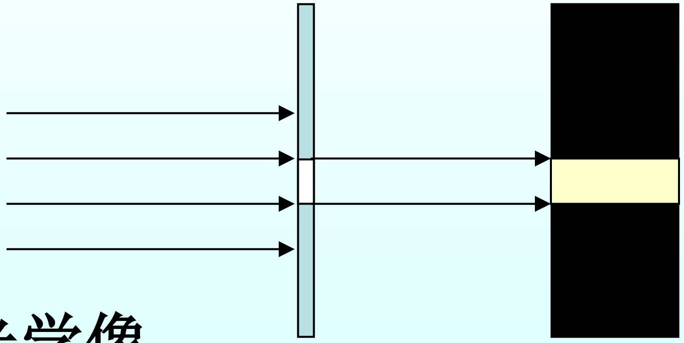
Discussion:

(1) $\Delta x \propto \lambda$ 波长越长，条纹宽度越宽；

（2） 缝宽变化对条纹的影响；

$$\Delta x = \frac{1}{2}\Delta x_0 = f\frac{\lambda}{a}$$ 缝宽越小，条纹宽度越宽

当 $\dfrac{\lambda}{a} \to 0$ 时， $\Delta x \to 0$

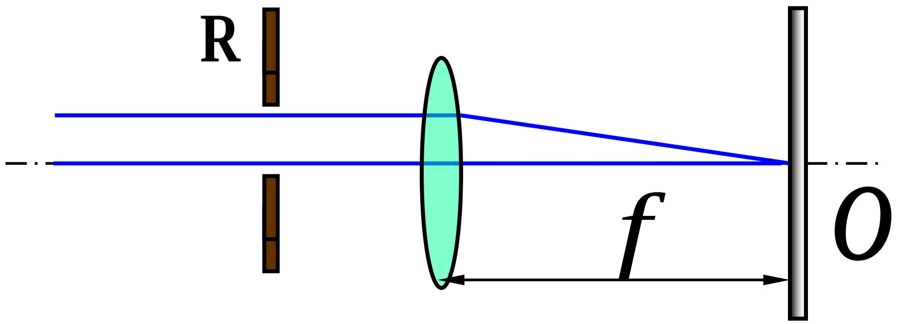


只显出单一的明条纹 ——单缝的几何光学像

∴几何光学是波动光学在 $\lambda/a \to 0$ 时的极限情形

(3) 单缝衍射的动态变化

◈ 单缝上下移动



单缝上移，零级明纹仍在透镜光轴上.

◈根据透镜成像原理衍射图不变 .

(4) 入射光非垂直入射时光程差的计算

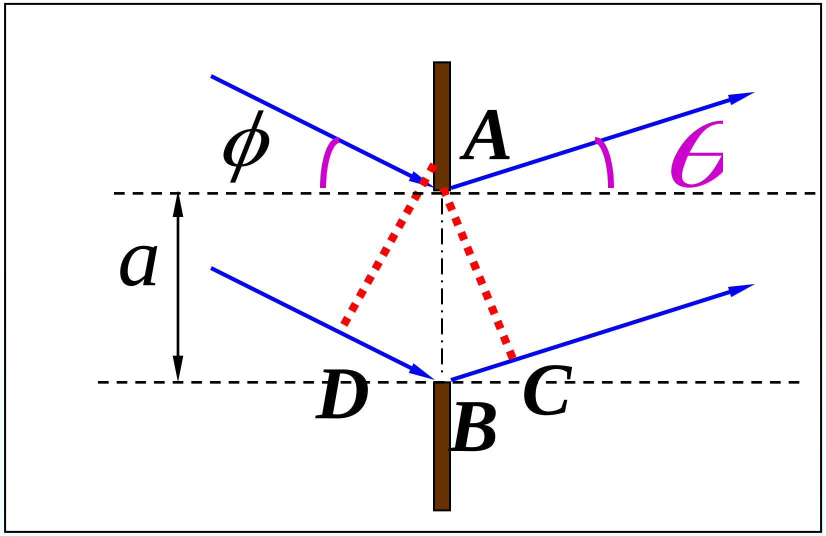
Monochromatic light falls on a transmission diffraction grating at an angle to the angle ϕ

$$\varDelta = DB + BC$$

$$= a(\sin\phi + \sin\theta)$$

$$= m_{\max}\lambda$$

$$m_{\max} = \frac{a(\sin\phi + 1)}{\lambda}$$



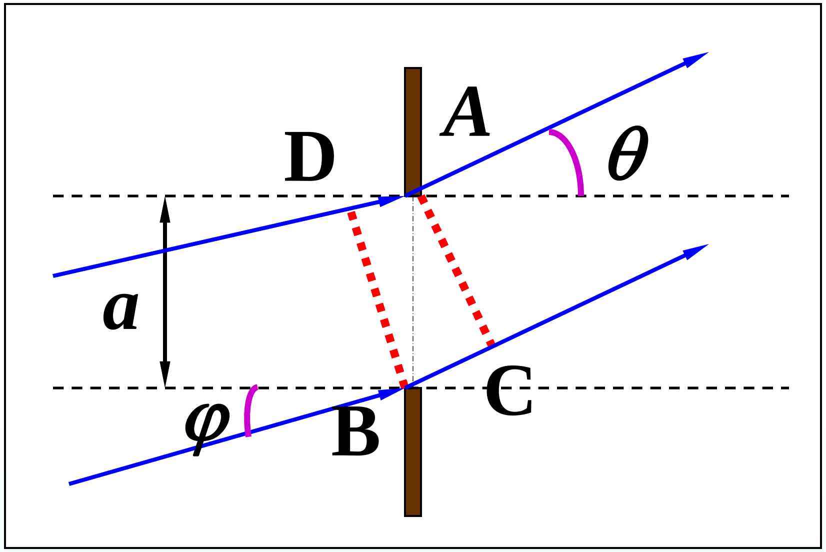
（中央明纹向下移动）

$$\Delta = a(\sin\phi + \sin\theta) = m_{\min}\lambda$$

$$m_{\min} = \frac{a(\sin\phi - 1)}{\lambda}$$

$$\Delta = BC - DA$$

$$= a(\sin\theta - \sin\phi)$$



（中央明纹向上移动）

Example **When violet light of wavelength 415nm falls on a single slit, it creates a central diffraction peak that is 9.2cm wide on the screen that is 2.50m away. How wide (mm) is this slit?**

Solution:

$$\Delta x_0 = f\,\Delta\theta_0 = 2f\frac{\lambda}{a}$$

$$a = 2f\frac{\lambda}{\Delta x_0}$$

$$= 2\times 2.50\times\frac{415\times 10^{-9}}{9.2\times 10^{-2}} = 2.25\times 10^{-5}m = 0.023mm$$

Example $a' = \frac{1}{2}a$

焦平面上原来3 级暗纹处，现在明暗情况如何

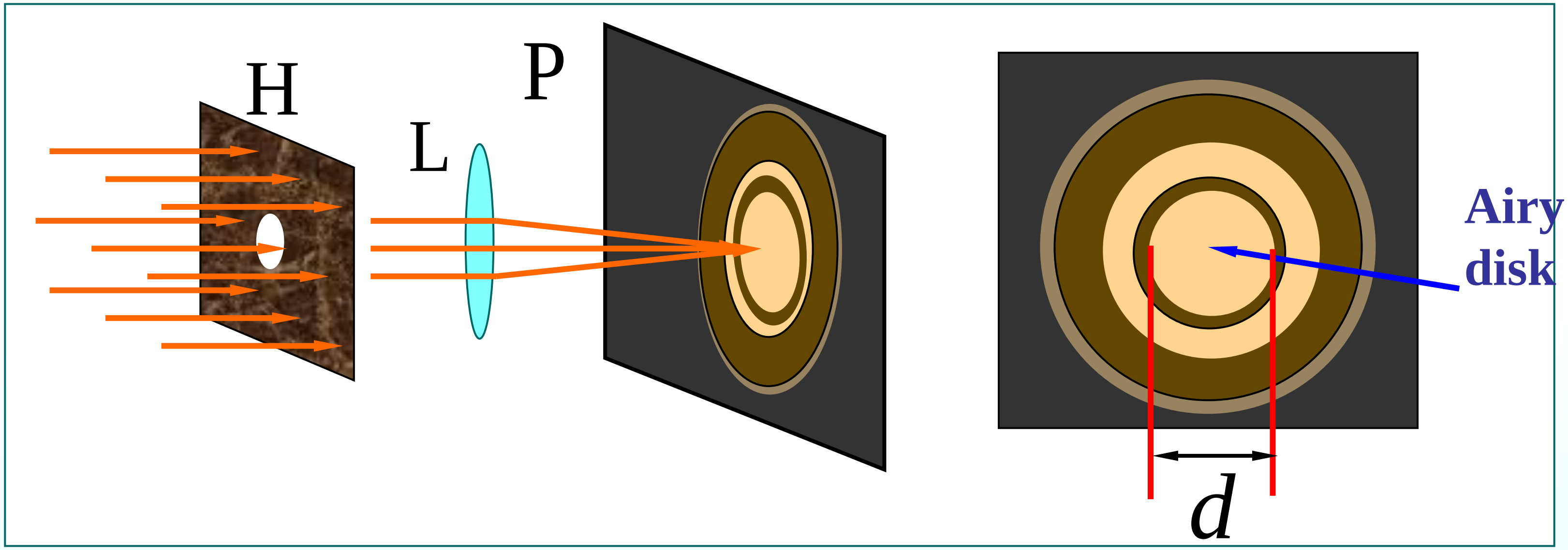
$$a\sin\theta = 3\lambda$$

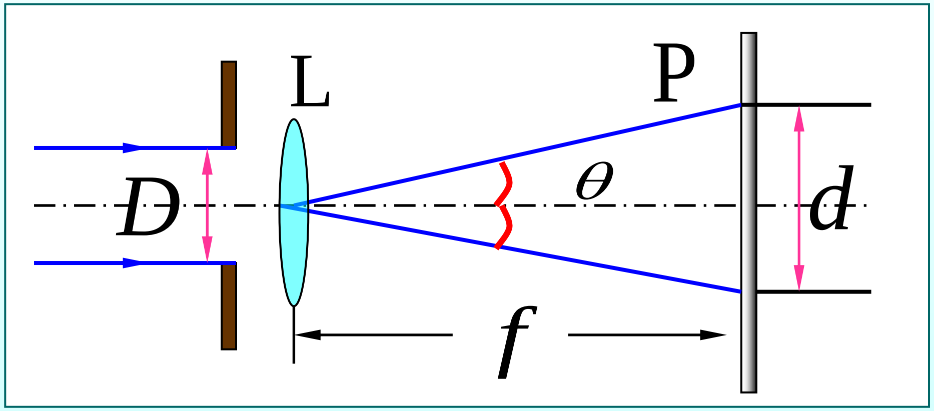
$$\frac{1}{2}a\sin\theta = \frac{3}{2}\lambda = (2m'+1)\frac{1}{2}\lambda \Rightarrow m' = 1$$

the first order of bright band

19.3 Limits of Resolution; Circular Apertures

1. The diffraction of circular apertures





d: diameter of Airy disk

D: diameter of circular opening

$$\theta = \frac{\lambda}{D/1.22} = 1.22\frac{\lambda}{D} = \frac{d/2}{f}$$

Angular half width

2. Resolution

The ability that an optical devise can just distinct two points being very close together.

几何光学与波动光学的区别

几何光学 ：

(经透镜)

物点 ⇒ 象点

不考虑艾里斑

物(物点集合) ⇒ 象(象点集合)

波动光学 ：

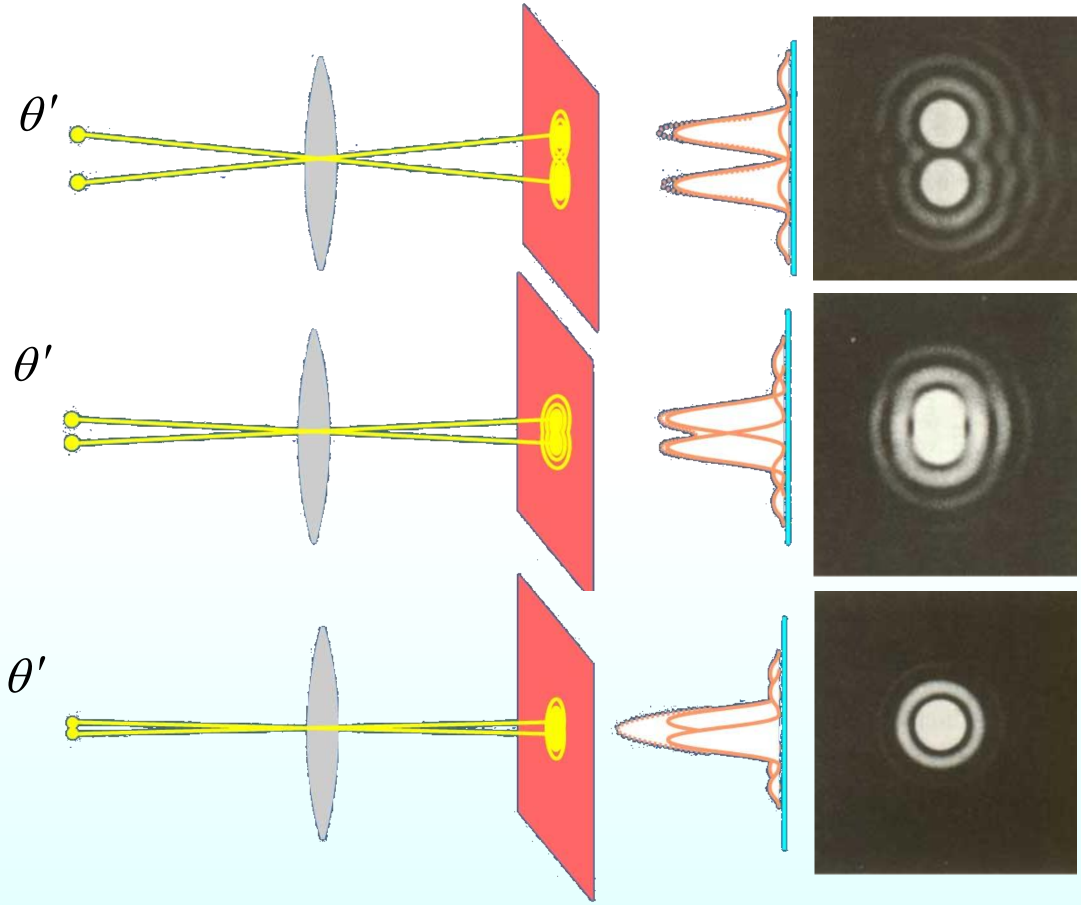
（ 经透镜 ）

物点 ⇒ 象斑

物(物点集合) ⇒ 象(象斑集合)

考虑艾里斑

Rayleigh criterion



$\theta' > \theta$

resolvable

$\theta' = \theta$

resolvable

$\theta' < \theta$

not resolvable

Two images are just resolvable when the center of the diffraction disk of one image is directly over the first minimum in the diffraction pattern of the other.

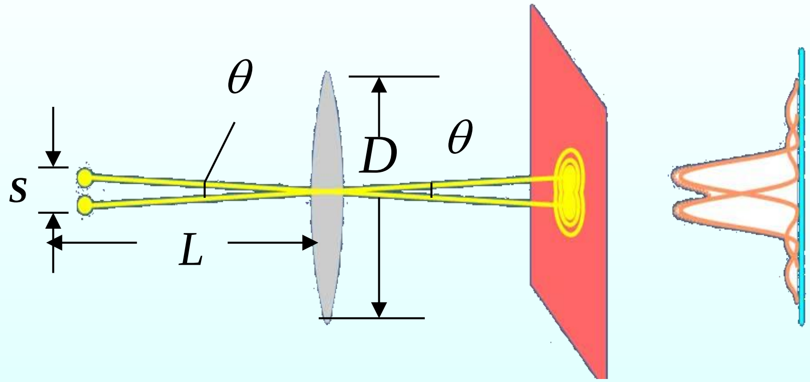
Angle of minimum resolution

$$\theta = 1.22\frac{\lambda}{D}$$

Resolution of optical devise 分辨本领

$$R = \frac{1}{\theta} = \frac{D}{1.22\lambda} \propto D, \quad \frac{1}{\lambda}$$

$$s = L\theta = 1.22\frac{\lambda L}{D}$$



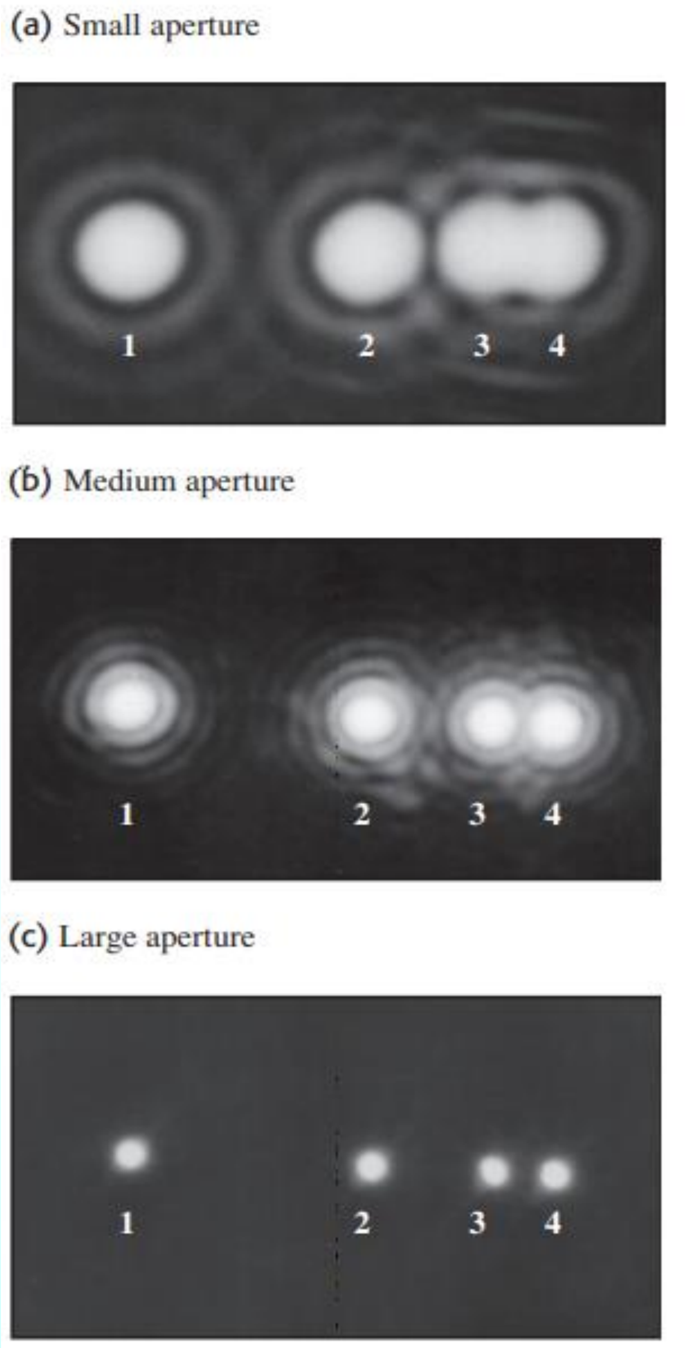
$$If \quad f \simeq \frac{D}{2} \qquad RP = s \simeq \frac{\lambda}{2}$$

It is not possible to resolve detail of objects smaller than the wavelength of the radiation being used.

$$R = \frac{1}{\theta} = \frac{D}{1.22\lambda} \propto D, \quad \frac{1}{\lambda}$$

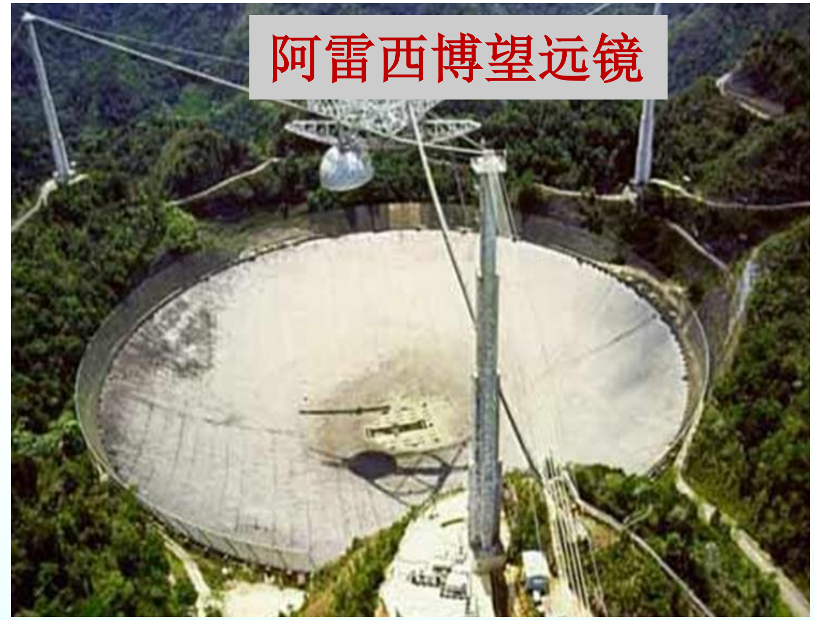
Rayleigh's criterion shows that resolution improves with larger diameter; it also improves with shorter wavelengths.

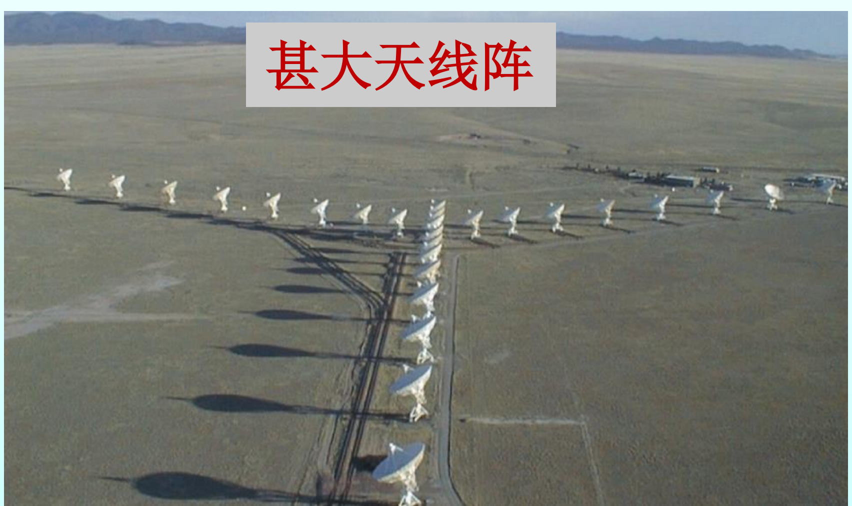
$$\lambda = \frac{h}{p} = \frac{h}{m_0 v} = \frac{h}{\sqrt{2em_0U}}$$

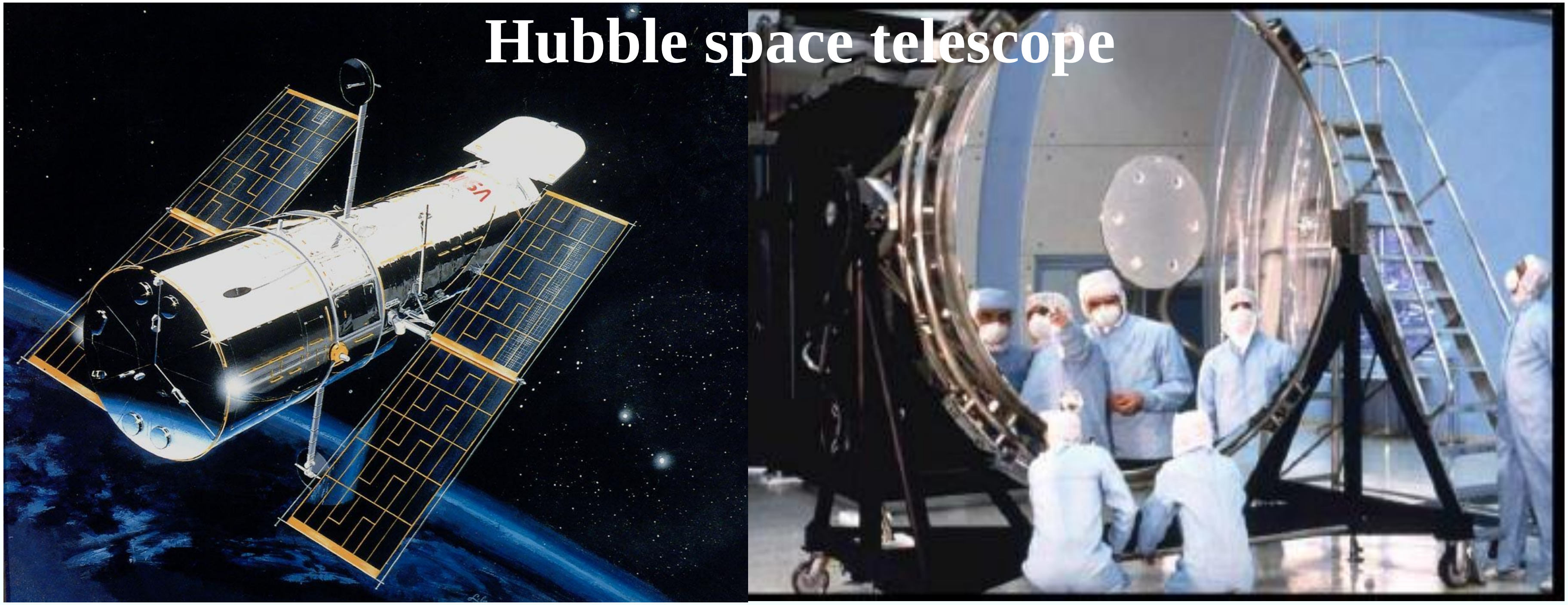


应用：

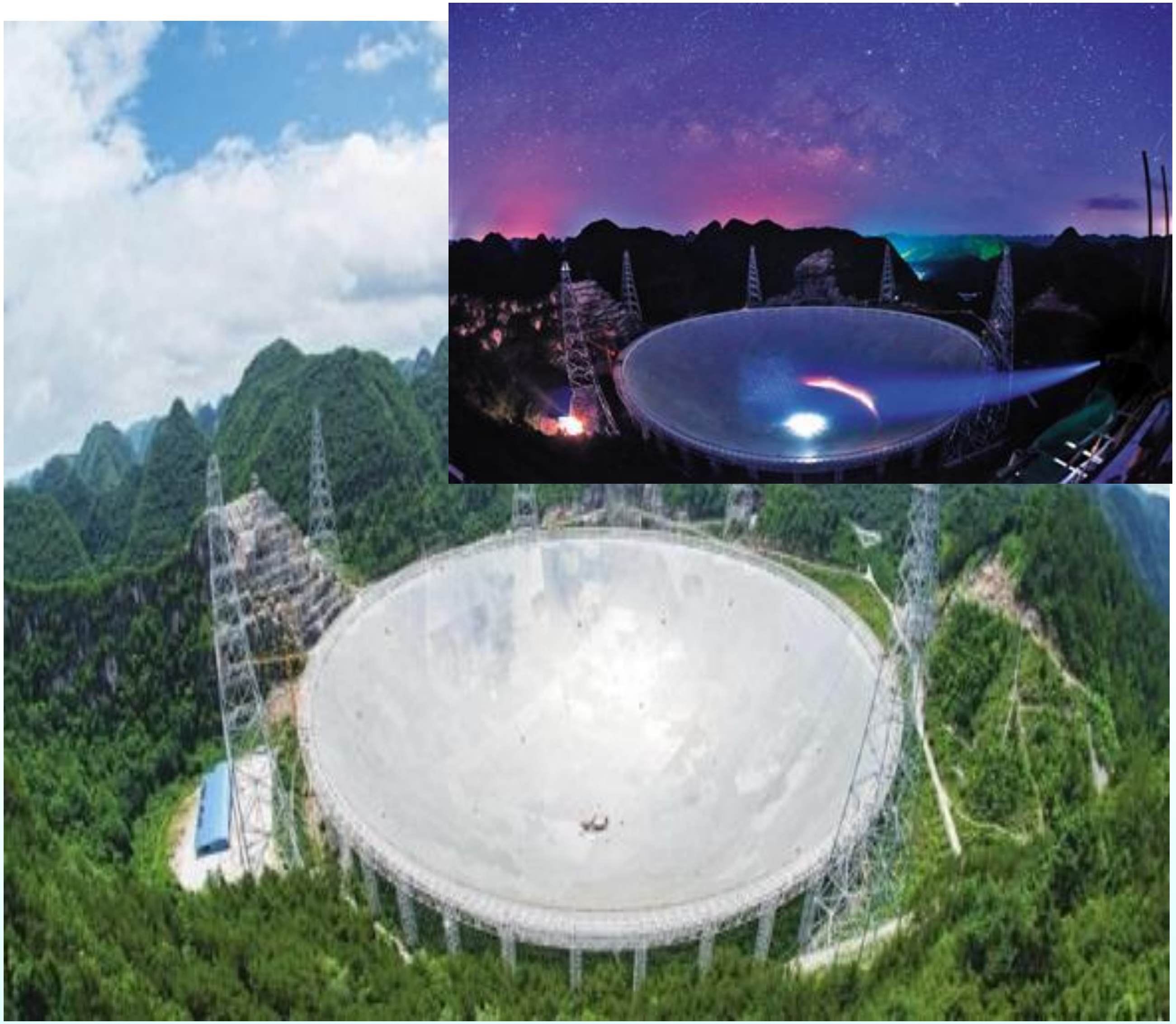
望远镜：λ 不可选择，可 $\uparrow D \rightarrow \uparrow R$







1990 年发射的哈勃太空望远镜的凹面物镜的直径为2.4 m，最小分辨角$\boldsymbol{\theta_0 = 0.1''}$在大气层外 615 km 高空绕地运行, 可观察130亿光年远的太空深处，发现了500亿个星系 .



FAST是世界在建的最大射电望远镜，借助天然圆形溶岩坑建造。FAST的反射镜边框是

中国天眼**:500米口径球面射电望远镜（Five-hundred-meter Aperture Spherical radio Telescope）FAST。**

应用：

显微镜：　D不会很大，　　可$\downarrow\lambda\rightarrow\uparrow R$

电子λ：　（10^{-2}~10^{-1} nm）

电子显微镜的分辨本领比光学显微镜的高的多

光学显微镜

电子显微镜

Resolution of the Eye

Estimate the limiting angle of resolution for the human eye, assuming its resolution is limited only by diffraction.

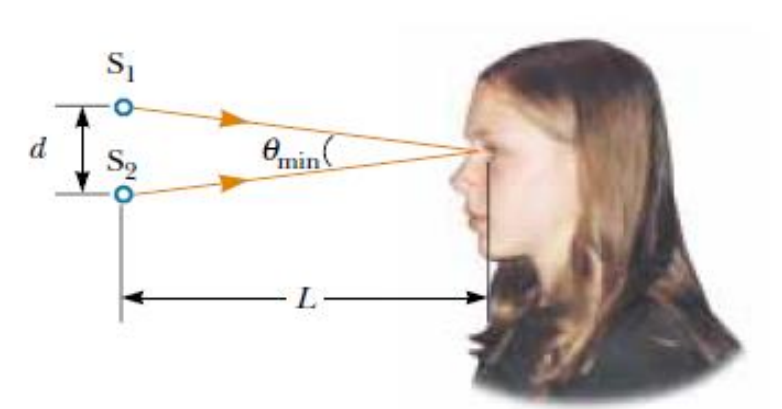
Solution: Let us choose a wavelength of 500 nm, near the center of the visible spectrum and estimate a diameter of 2.0 mm.

$$\theta = 1.22\frac{\lambda}{D} = \frac{(1.22)(500\times10^{-9})}{2\times10^{-3}} = 3\times10^{-4}\,rad$$

if the point sources are 25 cm from the eye

$$d \approx L\cdot\theta = (25cm)(3\times10^{-4}\,rad) = 8\times10^{-3}\,cm$$

This is approximately equal to the thickness of a human hair.

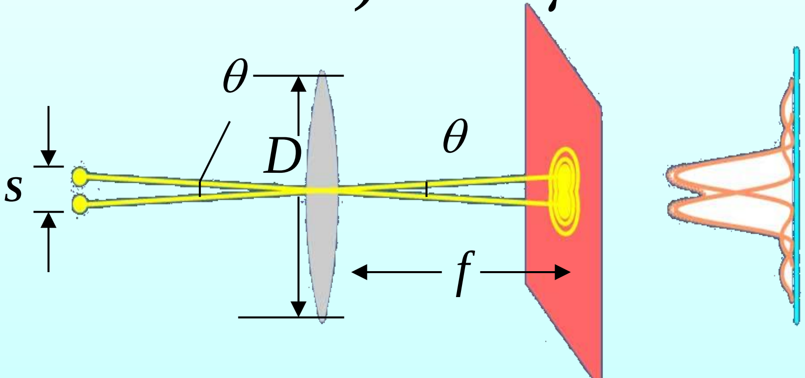


Example A circular converging lens, with diameter d=32mm and focal length f=24cm, form images of distant point objects in the focal plane of the lens. The wavelength is λ=550nm. (a) What angular separation must two distant point objects have to satisfy Rayleigh's criterion? (b) What is the separation Δx of the centers of the images in the focal plane?

Solution:

$$(a)\quad \theta = 1.22\frac{\lambda}{D} = \frac{(1.22)(550\times10^{-9})}{32\times10^{-3}} = 2.1\times10^{-5}\,rad$$

$$(b)\quad \Delta x \approx f\cdot\theta = (0.24m)(2.1\times10^{-5}\,rad) = 5.0\mu m$$



Example **It is said that a camera in a spy-satellite can distinguish clearly the number of car's license. To distinguish the distance of 5cm between numbers on a license from the height of 160km, what should the diameter of camera be? (assume λ= 500nm)**

(要识别牌照上的字划间的距离为5cm. 在160km高空的照相机的孔径需要大)?

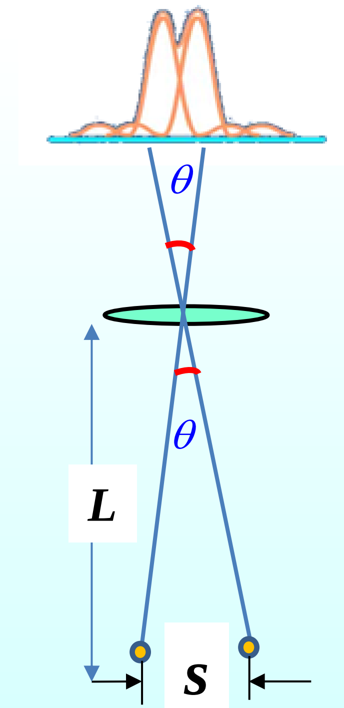
Solution: $s = 5\ \text{cm} = 0.05\text{m}$

$$\lambda = 500\ nm = 5.0\times10^{-7}\ m$$

$$L = 160\ \text{km} = 1.6\times10^{5}\ \text{m}$$

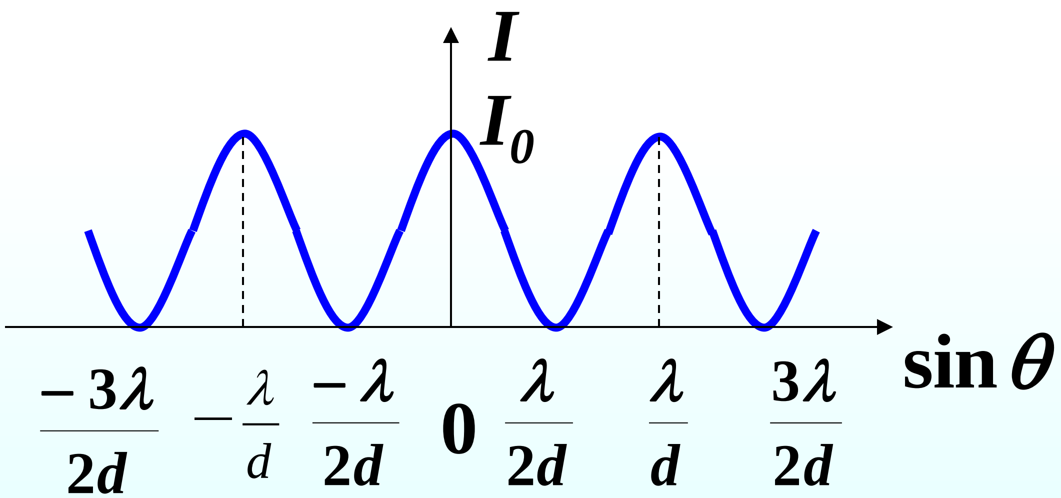
$$\theta = 1.22\frac{\lambda}{D} \qquad s \approx L\cdot\theta$$

$$D = \frac{1.22\lambda\cdot L}{s} = \frac{1.22\times5.0\times10^{-7}\times1.6\times10^{5}}{0.05} = 1.952\,\text{m}$$



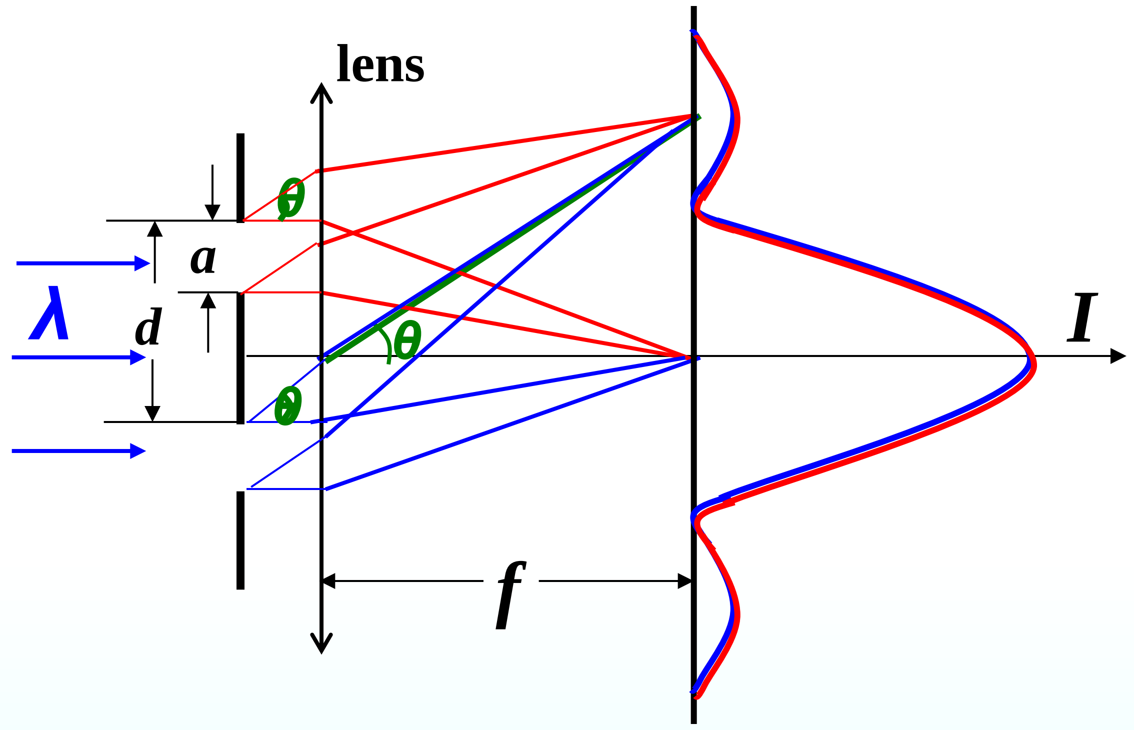
19.4 Diffraction in the Double-slit Experiment

Intensity of the Double-slit Experiment



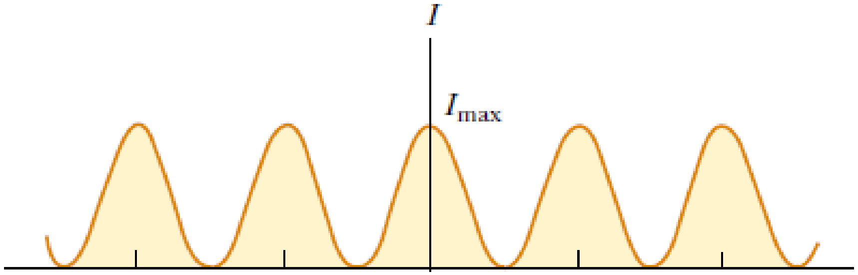
constructive

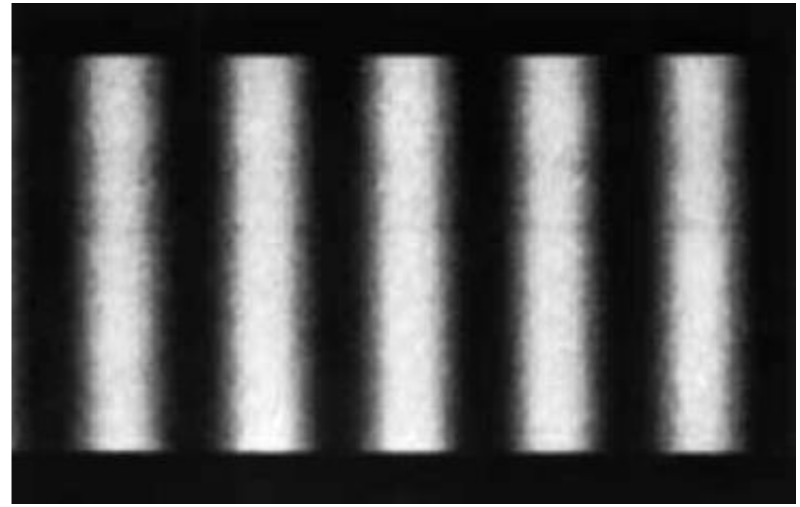
destructive



设双缝的每个缝宽均为a，在夫琅禾费衍射下，每个缝的衍射图样位置是相重叠的。

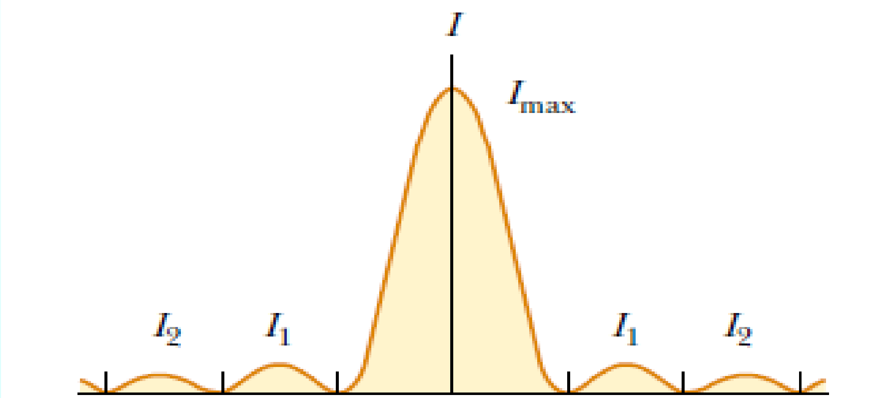
$$I_\theta = I_0 \cos^2\left(\frac{\pi d}{\lambda}\sin\theta\right)$$





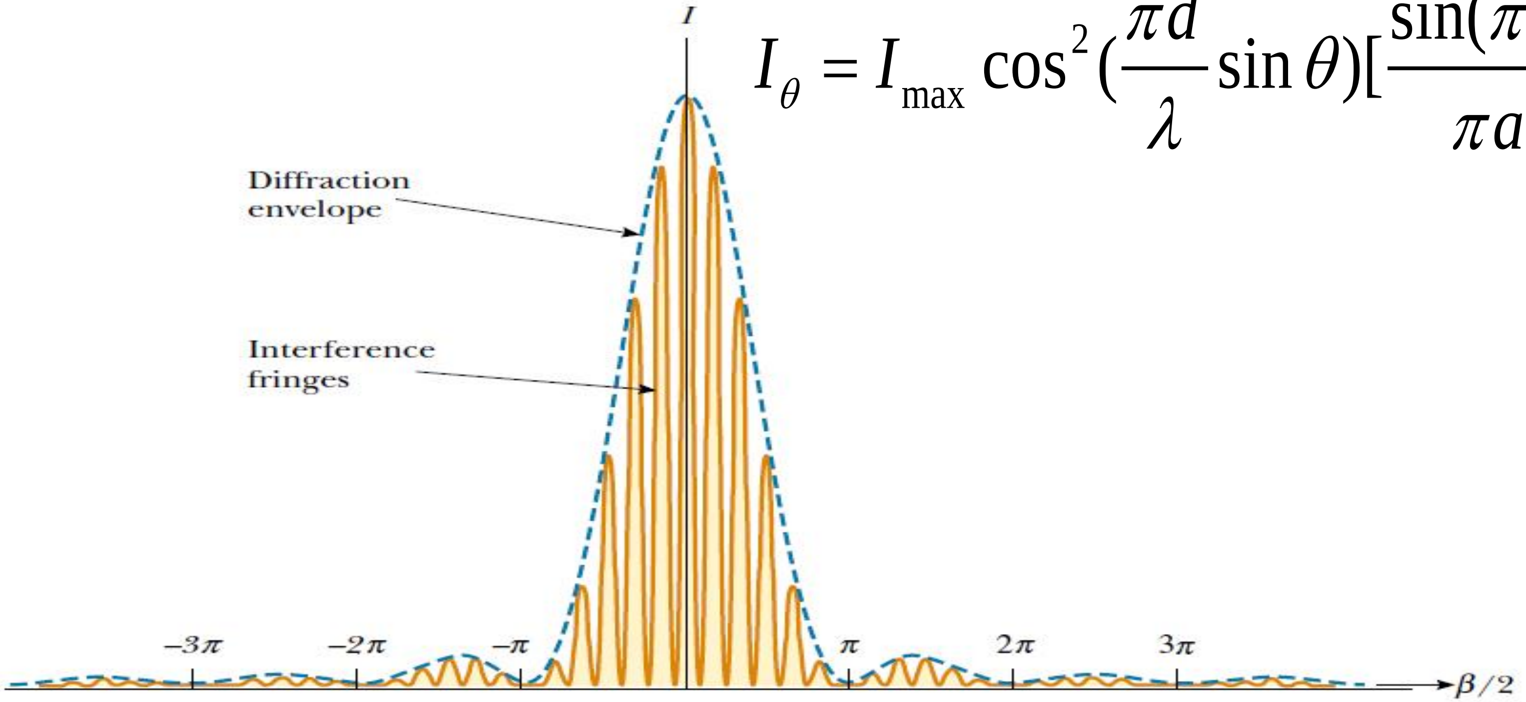
Double-slit interference

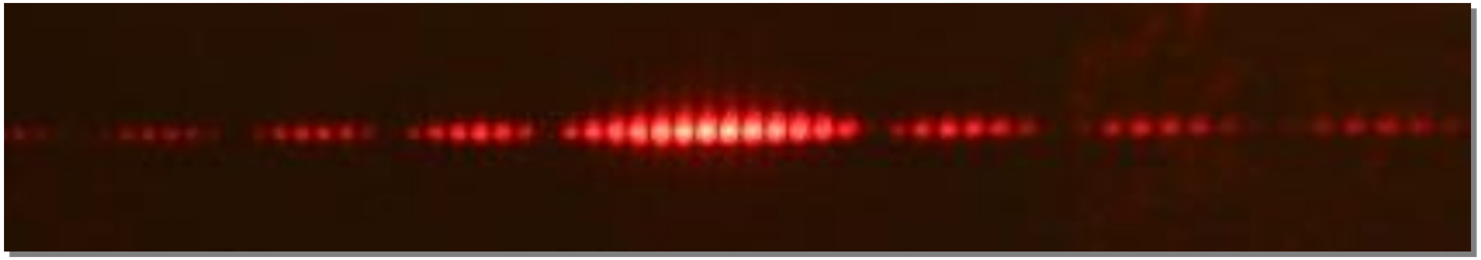
$$I_\theta = I_0\left[\frac{\sin(\pi a \sin\theta/\lambda)}{\pi a \sin\theta/\lambda}\right]^2$$



Single-slit diffraction

$$I_\theta = I_{\max}\cos^2\left(\frac{\pi d}{\lambda}\sin\theta\right)\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right]^2$$





$$\frac{d}{a} = 6$$

The combined effects of diffraction and interference. The diffraction pattern acts as an "envelope" and controls the intensity of the regularly spaced interfere maximum.

Destructive

Double-slit interference

$$d\sin\theta = \pm(2m+1)\frac{\lambda}{2}\cdots m = 0,1,2\cdots \min$$

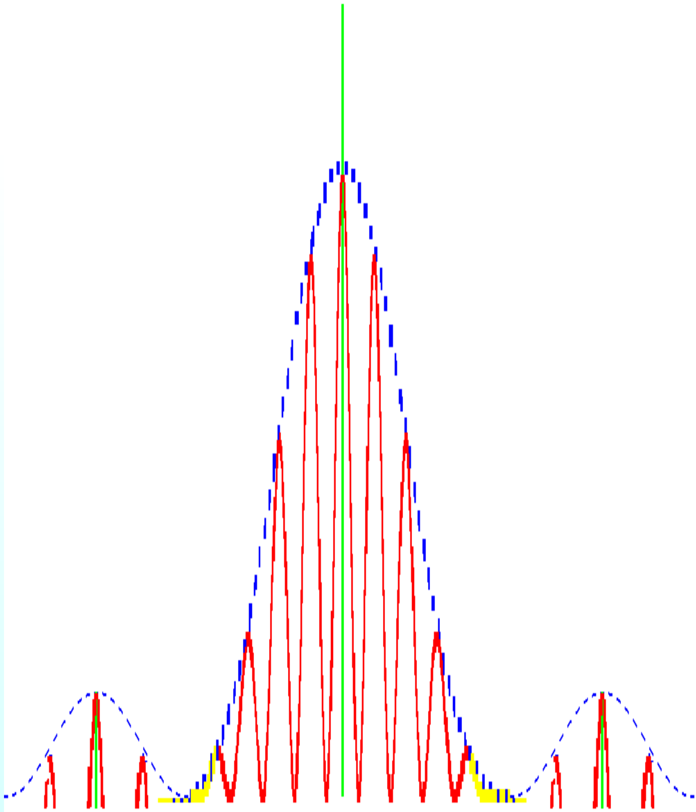
One-slit diffraction

$$a\sin\theta = \pm m\lambda\cdots m = 1,2\cdots \min$$

Constructive

Double-slit interference

$$d\sin\theta = \pm m\lambda\cdots m = 0,1,2\cdots \quad max$$

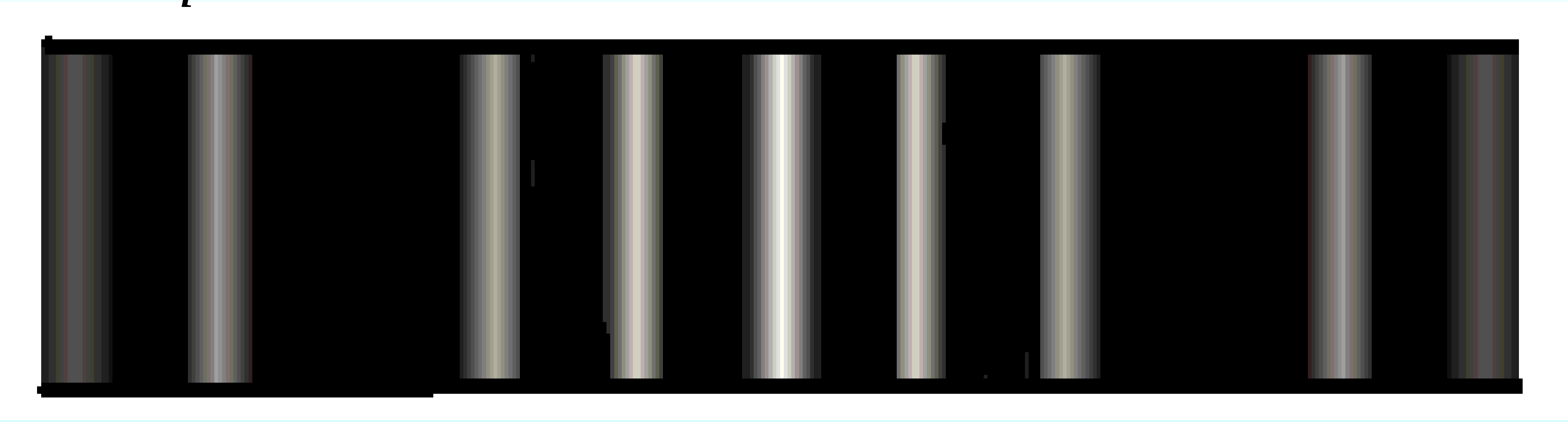


$$\begin{cases} a\sin\theta = m\lambda \cdots\cdots m = \pm 1, \pm 2 \cdots\cdots \min \\ d\sin\theta = m'\lambda \ldots\ldots m' = 0, \pm 1, \pm 2 \cdots\cdots \max \end{cases}$$

强度为零的衍射光相干，相长干涉的强度仍为零—缺级

Missing order: $\dfrac{d}{a} = \dfrac{m'}{m} = \dfrac{p}{q}$ ***p*, *q***为互质整数比时，缺 ***np*** 级

$\dfrac{m'}{m} = \dfrac{p}{q} = 3\left(or\ \dfrac{3}{2}\right) \Rightarrow$ 缺第 **3** ， **6**、**9**、**12** …… 级



Question: **Design a double-slit apparatus so that the central diffraction peak contains precisely fifteen fringes.**

$$\frac{d}{a} = 8$$

Example :In a double-slit diffraction experiment, the wavelength λ of the light source is 405nm, the slit separation d is 19.44 μm, and the slit width a is 4.050 μ m. (a) How many bright interference fringes are within the central peak if the diffraction envelope? (b) How many bright fringes are within either of the first side peaks of the diffraction envelope?

Solution: **(a) The first minima in the diffraction pattern**

$$a \sin\theta = \lambda$$

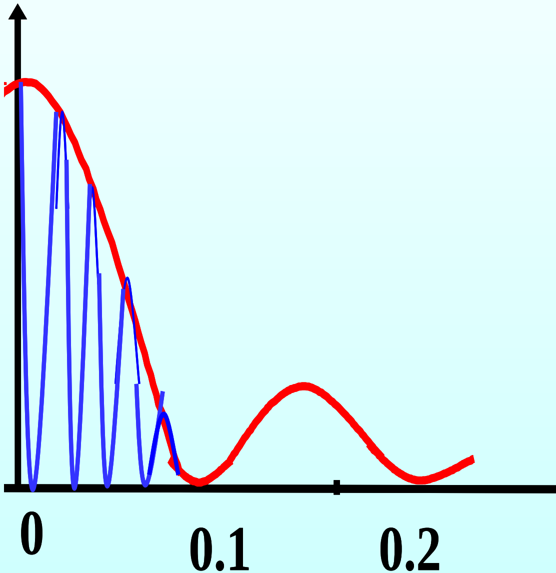
The angular locations of the bright fringes

$$d \sin\theta = m_2 \lambda$$

$$m_2 = \frac{d}{a} = 4.8 \qquad n = 2 \times 4 + 1 = 9$$

$$(b) \quad a \sin\theta = 2\lambda$$

$$m_2 = \frac{2d}{a} = 9.6 \qquad n = 9 - 4 = 5$$



In a double-slit diffraction(双缝衍射) experiment the number of interference fringes within the central diffraction maximum(中央衍射峰) can be increased by （多选）

(A)increasing the wavelength λ

(B)decreasing the wavelength λ

(C) decreasing the slit width a **(缝宽)**

(D)decreasing the slit separation d **(两缝之间距离)**

(E) increasing the slit width a

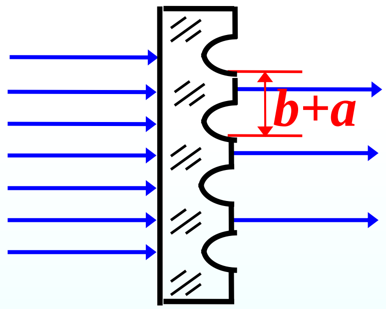
(F) increasing the slit separation d

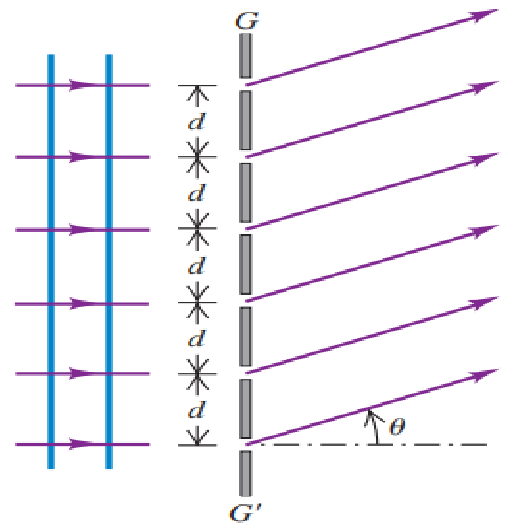
19.5 Diffraction Grating

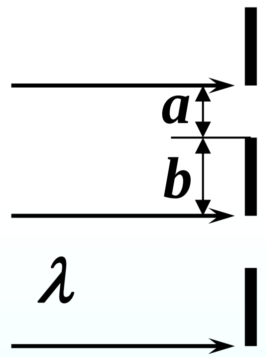
1.Grating: **a number of equally spaced parallel slits (rulings)**

(10^4 -10^5 lines per cm)

Transmission grating







d*: Grating spacing** (The separation between rulings***)

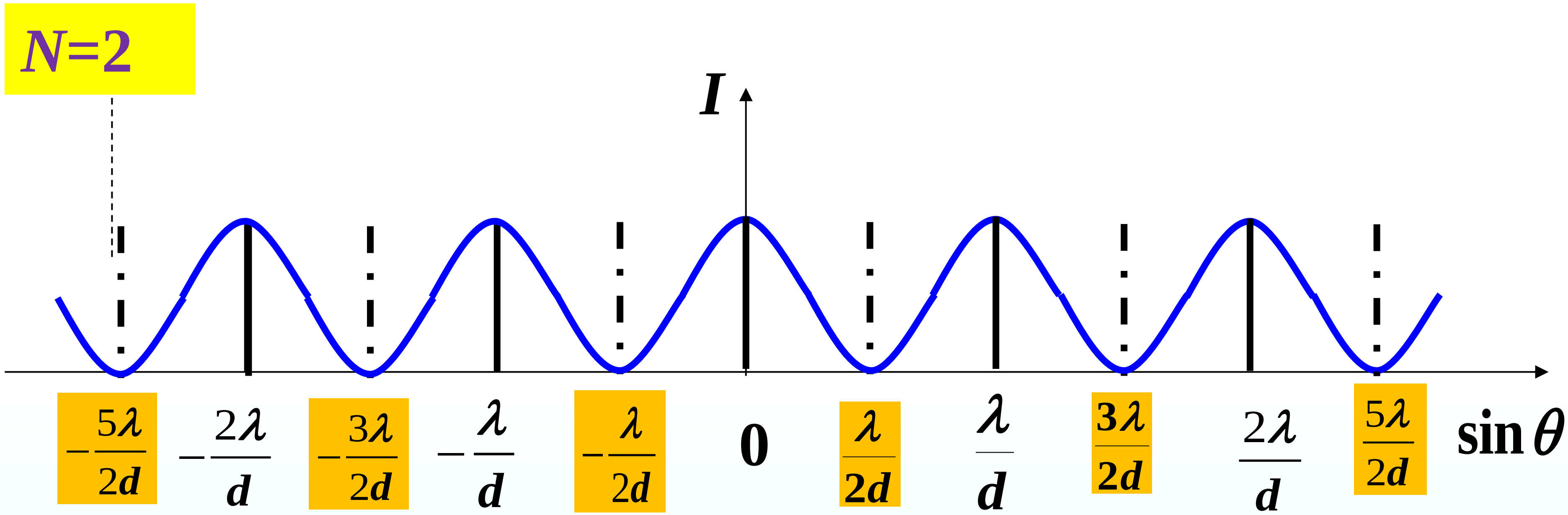
If N rulings occupy a total width w , then d =w/N

$$d = a + b \quad \text{光栅常数}$$

a: **the width of a slit**

b: **the width of the opaque part**

只考虑缝间干涉　$d = a + b$　$d\sin\theta = \pm m\lambda,\quad m = 0, 1, 2, \cdots$



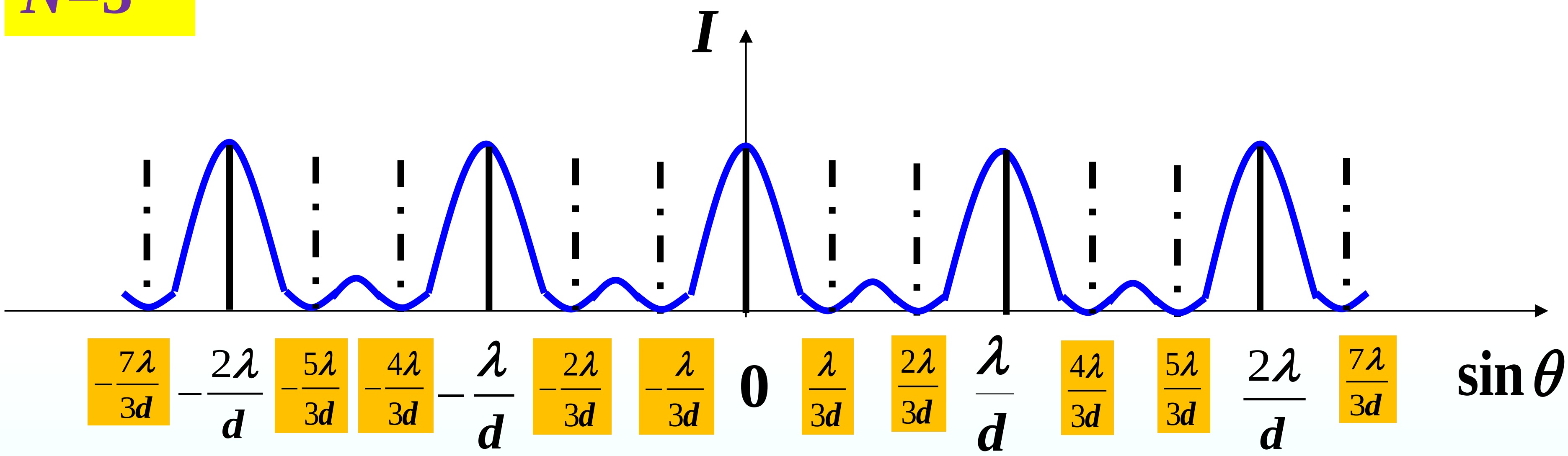
$d\sin\theta = \pm m\lambda,\quad m = 0, 1, 2, \cdots$ **constructive**

$d\sin\theta = \pm(2m+1)\frac{\lambda}{2},\quad m = 0, 1, 2, \cdots$ **destructive**

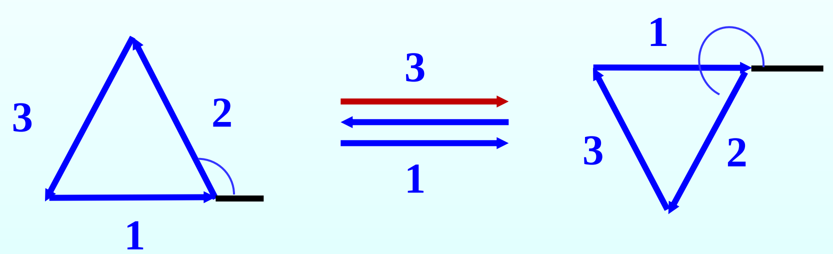
只考虑缝间干涉

$d = a + b \qquad d\sin\theta = \pm k\lambda, \quad k = 0, 1, 2, \cdots$

***N*=3**



暗纹　　次明纹　　暗纹

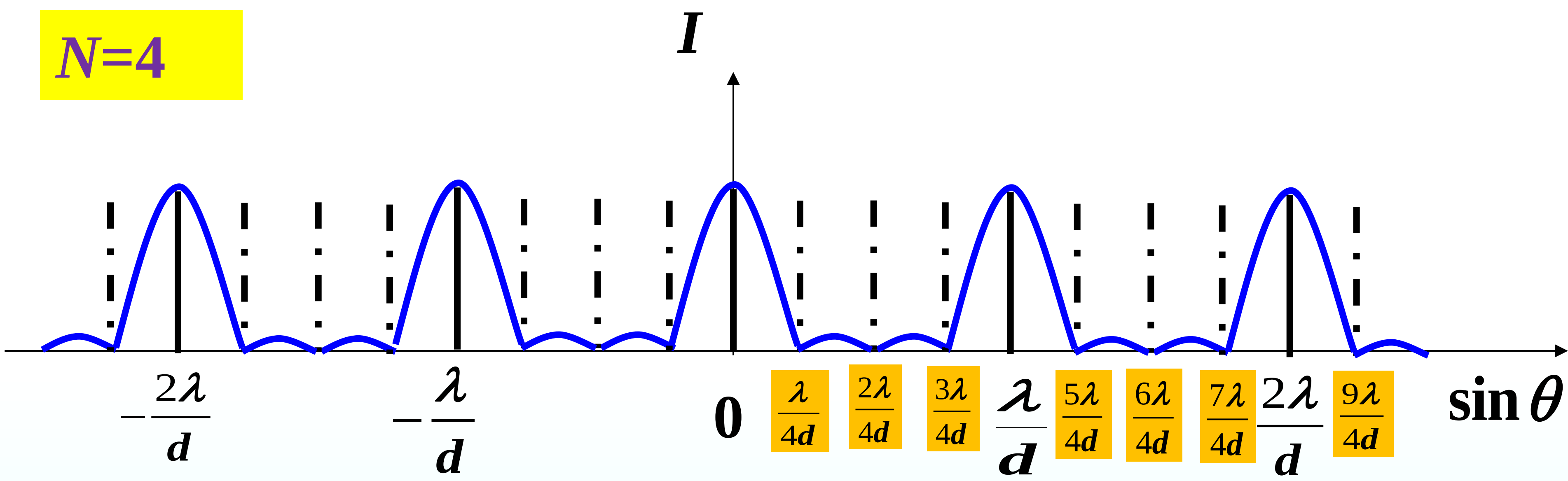


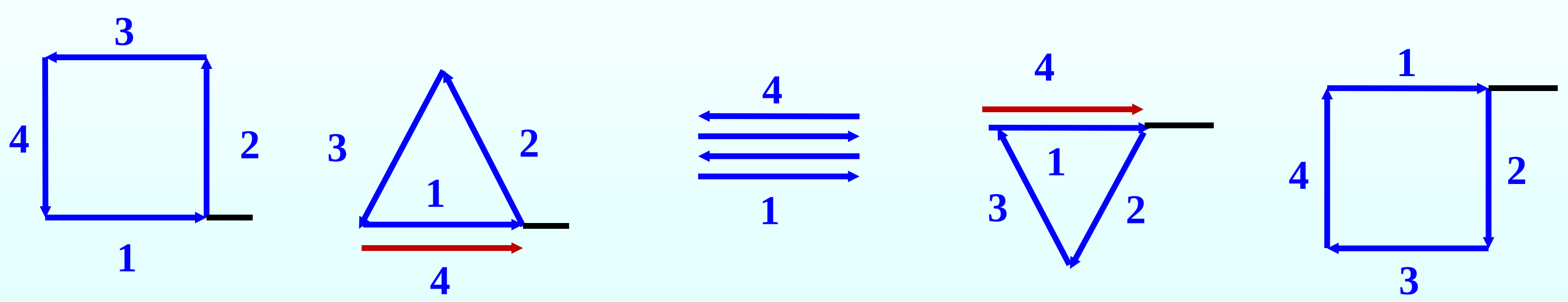
$\Delta\varphi = 2\pi/3 \qquad \Delta\varphi = \pi \qquad \Delta\varphi = 4\pi/3$

$\sin\theta = \frac{\lambda}{3d} \qquad \sin\theta = \frac{\lambda}{2d} \qquad \sin\theta = \frac{2\lambda}{3d}$

只考虑缝间干涉 $d=a+b$ $\quad d\sin\theta=\pm k\lambda,\quad k=0,1,2,\cdots$

N=4



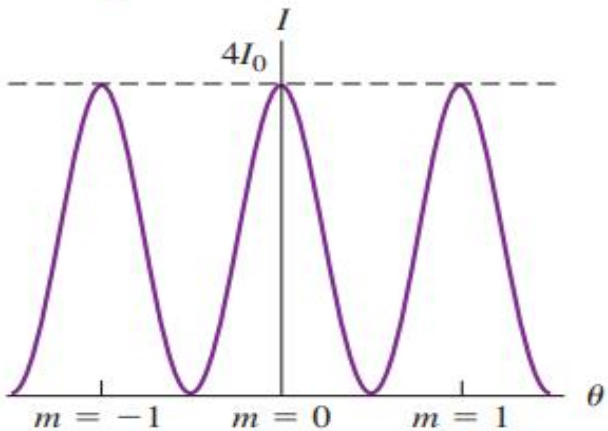


$\Delta\varphi=\pi/2$ $\quad \Delta\varphi=2\pi/3$ $\quad \Delta\varphi=\pi$ $\quad \Delta\varphi=4\pi/3$ $\quad \Delta\varphi=3\pi/2$

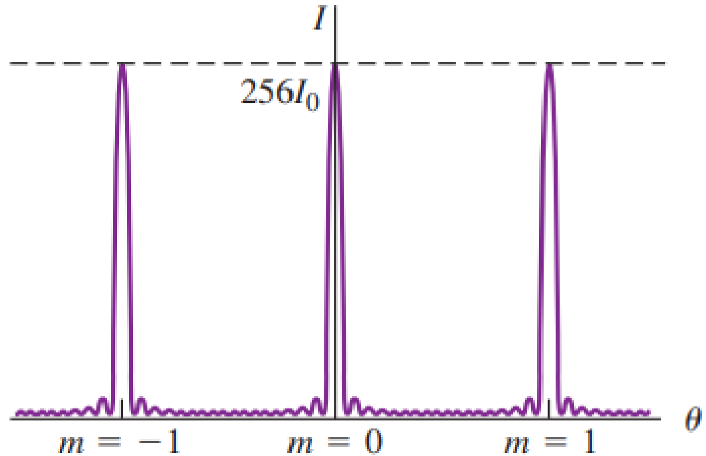
$\sin\theta=\frac{\lambda}{4d}$ $\quad \sin\theta=\frac{\lambda}{3d}$ $\quad \sin\theta=\frac{2\lambda}{4d}$ $\quad \sin\theta=\frac{2\lambda}{3d}$ $\quad \sin\theta=\frac{3\lambda}{4d}$

Interference patterns for equally spaced, very narrow slits.

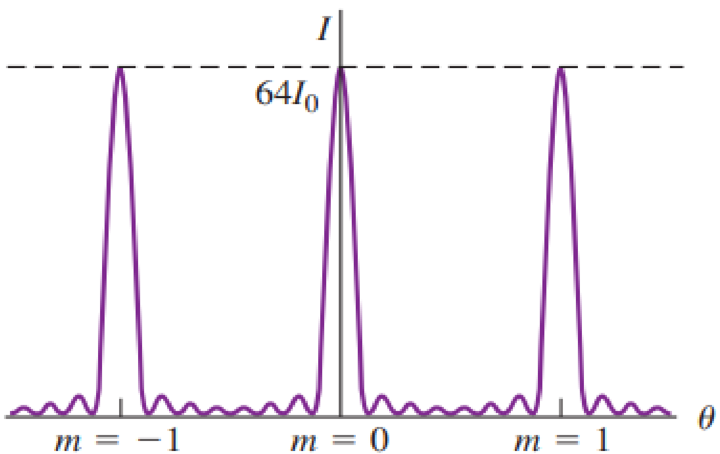
(a) $N = 2$: two slits produce one minimum between adjacent maxima.



(c) $N = 16$: with 16 slits, the maxima are even taller and narrower, with more intervening minima.



(b) $N = 8$: eight slits produce taller, narrower maxima in the same locations, separated by seven minima.

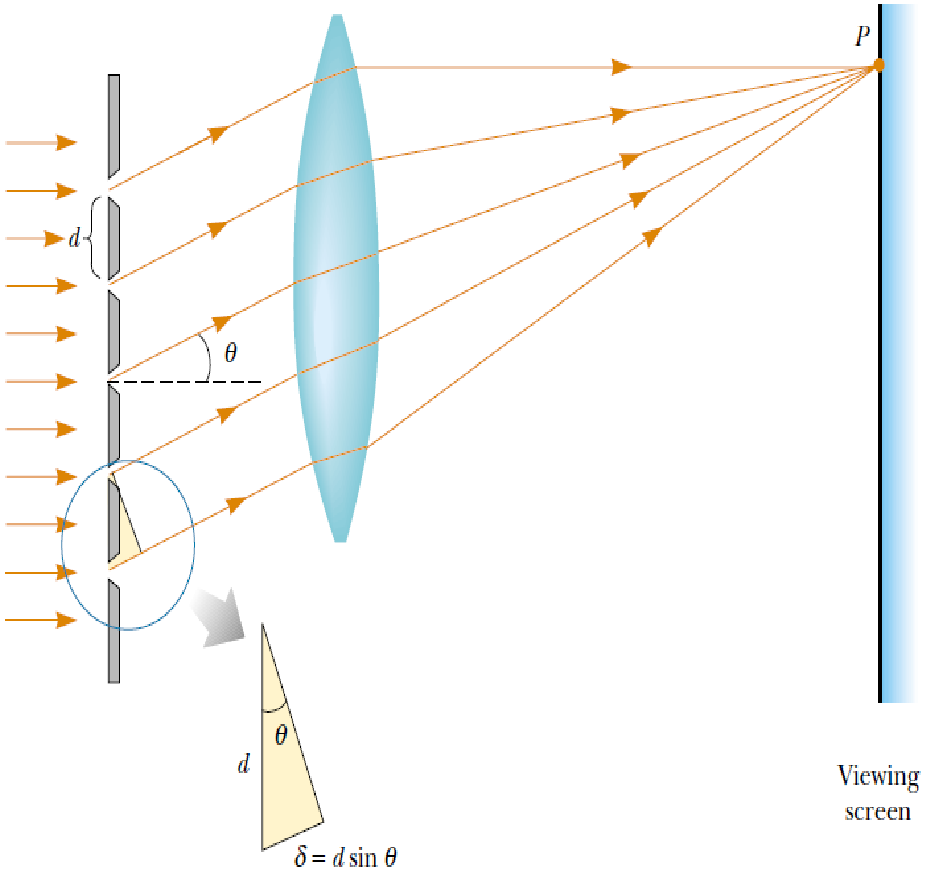


The greater the value of N, the narrower the principal maxima become. I_0 is maximum intensity for a single slit, and the maximum intensity for slits is $N^2 I_0$. The width of each peak (principal maximum) is proportional to $1/N$.

2. Grating equation

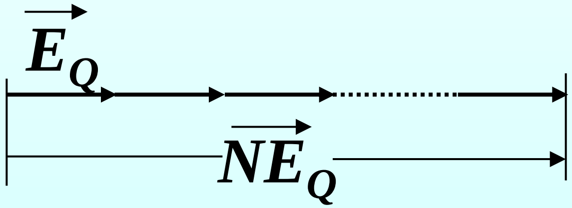
Grating diffraction= diffraction of every slit + interference lines (maxima) (明纹，主极大）

$$d\sin\theta=\pm m\lambda,\quad m=0,1,2\cdots$$



Conditions for interference maxima for a grating

maximum $\Delta\varphi=\pm 2k\pi$



$$I_Q\propto N^2E_Q^2$$

Dark fringe: **N个缝在Q点的N个振幅的矢量合为零**

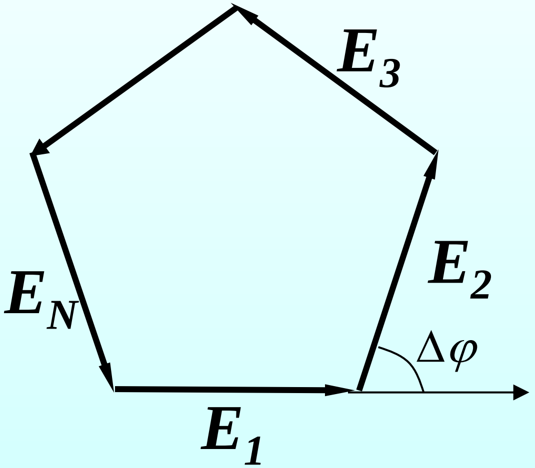
$$N\Delta\varphi = \pm 2m'\pi \qquad (1) \qquad \boldsymbol{m' = 1, 2, \cdots \neq Nk}$$

$$\Delta\varphi = \frac{(a+b)\cdot\sin\theta}{\lambda}\cdot 2\pi \quad (2)$$

Combine (1)、(2)

$$(m' \neq Nk, \quad m' \neq 0)$$

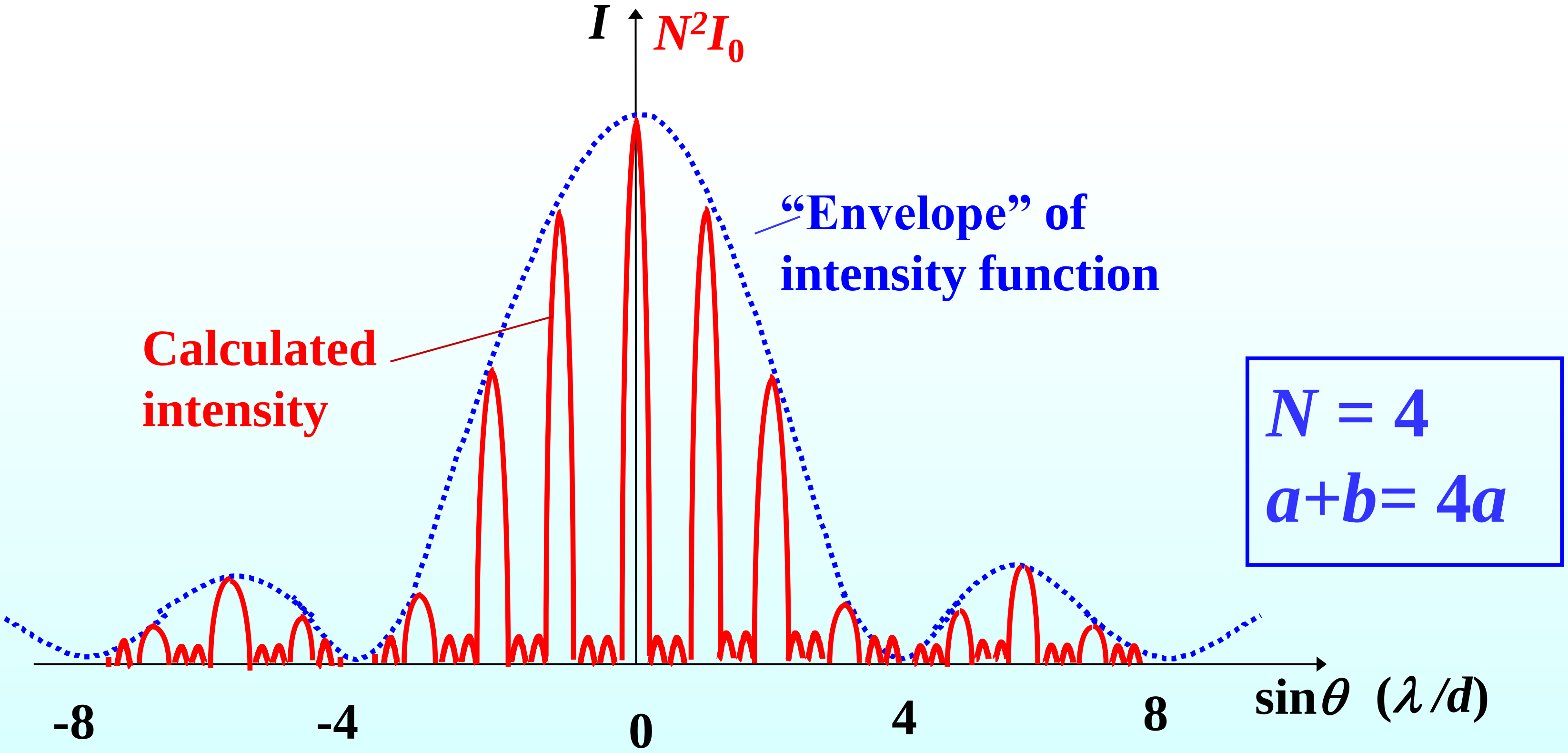
$$(a+b)\cdot\sin\theta = \pm\frac{m'}{N}\lambda$$



暗纹间距=(主明纹间距/N)

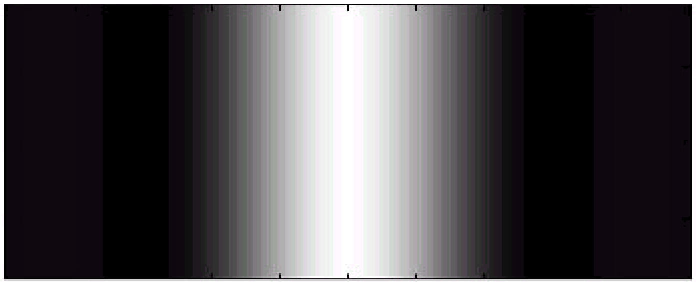
相邻主明纹间有 $N-1$ 个暗纹和 $N-2$ 个次明纹.

$$*I_\theta = I_0 \left[\frac{\sin(\pi a \sin\theta/\lambda)}{\pi a \sin\theta/\lambda}\right]^2 \left[\frac{\sin N(\pi d \sin\theta/\lambda)}{\sin(\pi d \sin\theta/\lambda)}\right]^2$$

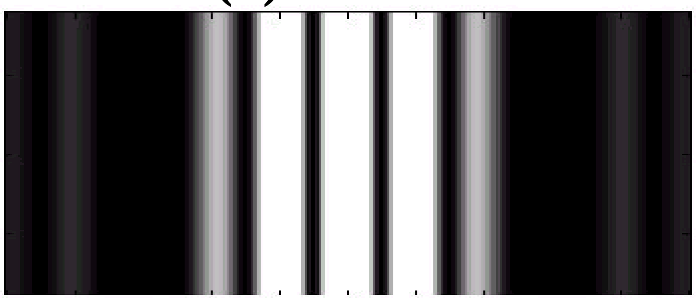


The greater of number of slits, the narrower and brighter the bright bands

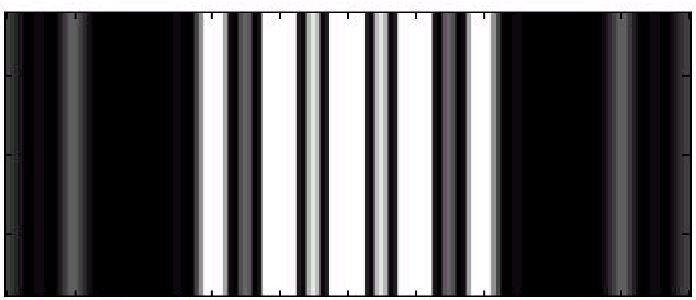
(a) 1 slit



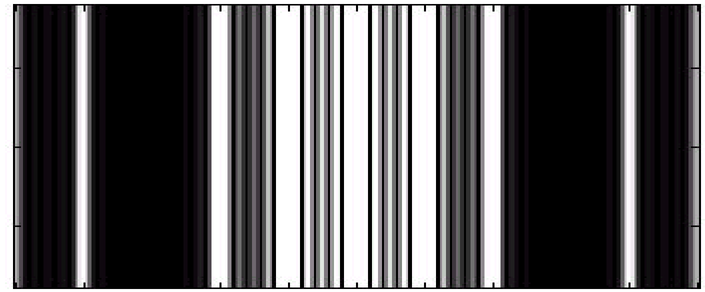
(b) 2 slits



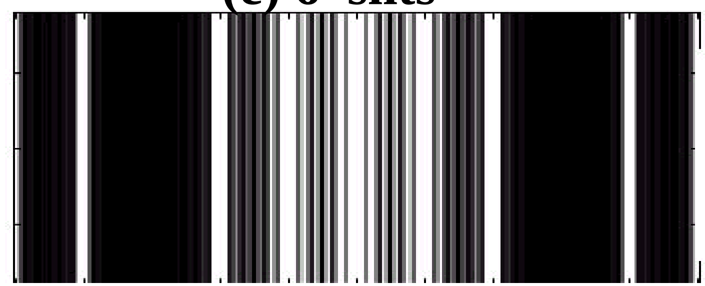
(c) 3 slits



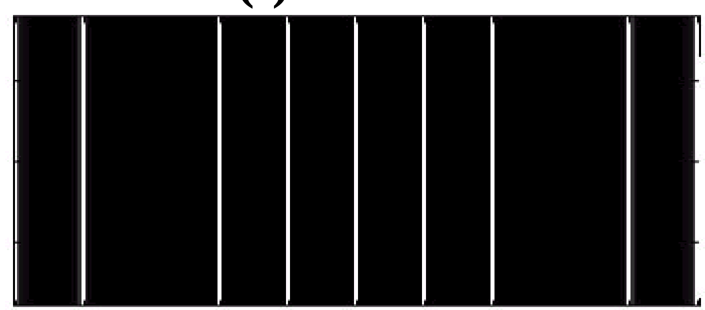
(d) 5 slits



(e) 6 slits



(f) 20 slits



3. 讨论:

(1) The maximum order 最高级次

$$d\sin\theta = m\lambda \qquad m = m_{\max} = \frac{d}{\lambda}\sin\theta \le \frac{d}{\lambda}$$

$$m_{\max} = [\frac{d}{\lambda}]$$

Number of bright lines=2m_{max}+1

(2) missing orders 缺级

$$a\sin\theta = m\lambda \cdots\cdots m = \pm 1, \pm 2 \cdots\cdots \min$$

$$d\sin\theta = m'\lambda \cdots\cdots m' = 0, \pm 1, \pm 2 \cdots\cdots \max$$

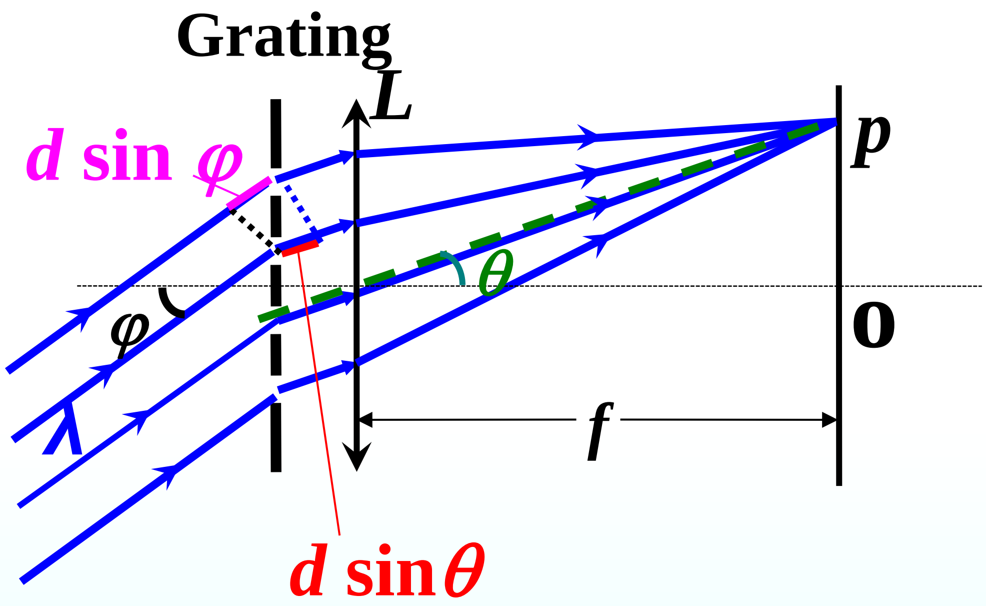
$$\frac{d}{a} = \frac{m'}{m} = \frac{p}{q}$$

***p*, *q* are integers， missing *np* orders**

Missing orders occur for a diffraction grating:

when a diffraction minimum coincides with an interference maximum.

(3) Light falls at a nonzero angle φ 斜入射

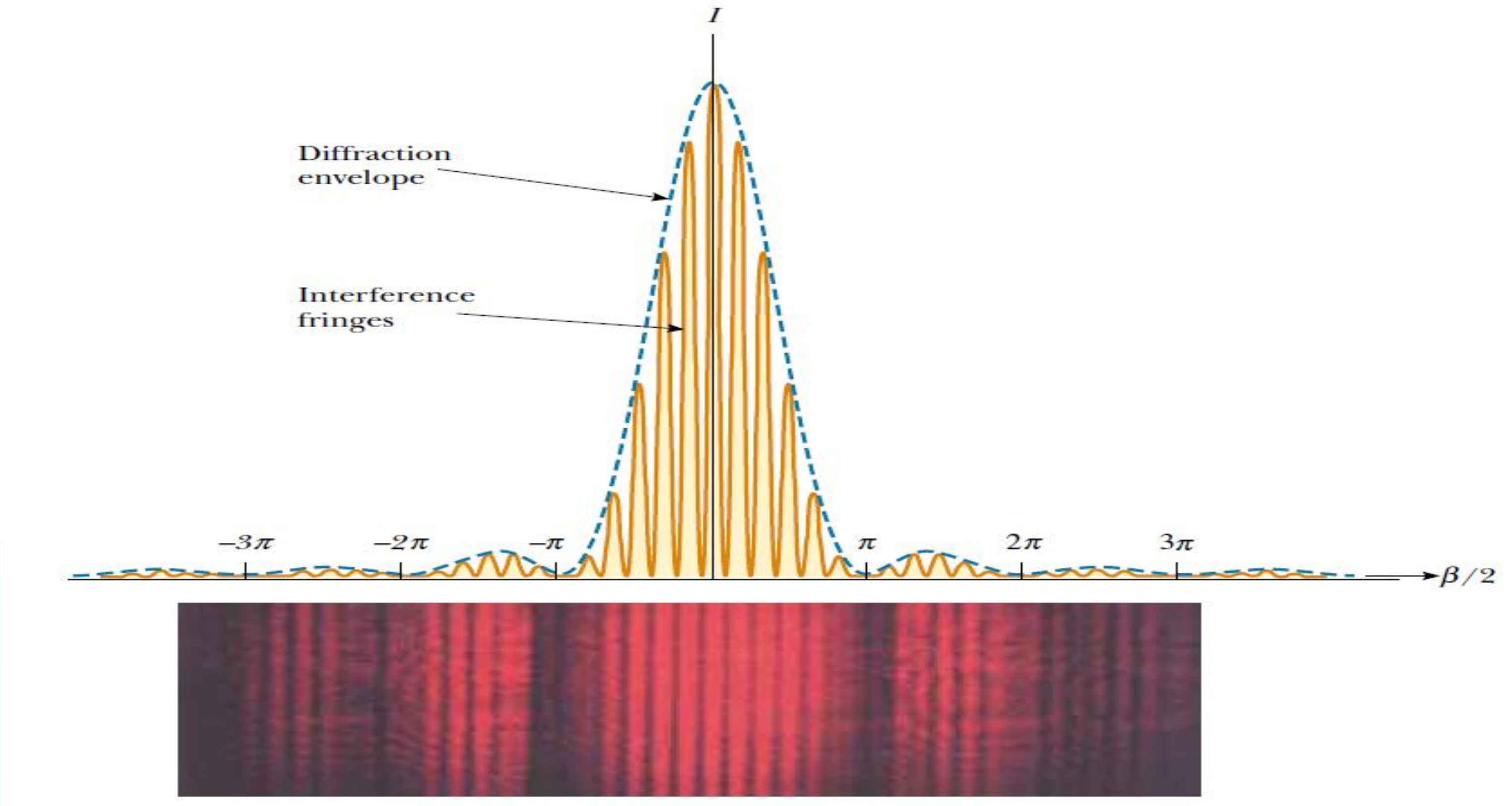


斜入射可以获得更高级次的条纹

$$\Delta = d(\sin\theta - \sin i)$$

$$\boxed{d(\sin\theta - \sin\varphi) = \pm m\lambda}$$ — diffraction grating

(4) Fringes in the central diffraction 中央衍射峰内的条纹数

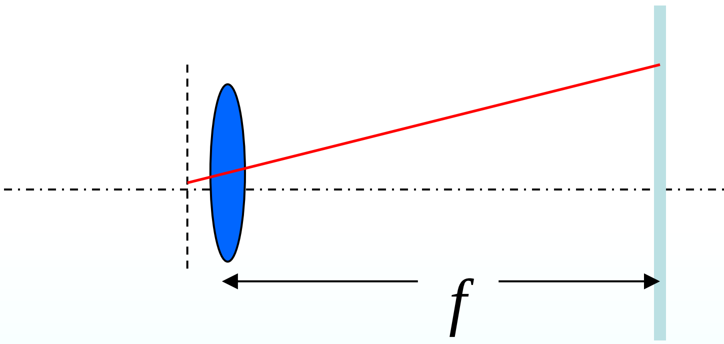


If $d/a = m$, where m is an integer, $N = 2m + 1 - 2 = 2m - 1$

If $d/a = m$, where m is a decimal number, $N = 2[m] + 1$

Example **Diffraction grating: Calculate the first- and second-order angles for light of wavelength 400 nm and 700nm if the grating contains10000 lines/cm.**

Solution: $$d = \frac{10^{-2}}{10000} = 1.0\times10^{-6}\, m$$



$$d \sin\theta = \pm m\lambda$$

In first order, the angles are

$$\sin\theta_{400} = \frac{m\lambda}{d} = \frac{1\times4.0\times10^{-7}}{1.0\times10^{-6}} = 0.400$$

$\sin\theta_{700} = 0.700$ **So** $\theta_{400} = 23.6^o$ $\theta_{700} = 44.4^o$

In second order

$$\sin\theta_{400} = \frac{m\lambda}{d} = \frac{2\times4.0\times10^{-7}}{1.0\times10^{-6}} = 0.800$$

$$\sin\theta_{700} = 1.40$$

So $\theta_{400} = 53.1^o$ θ_{700} (invisible)

Example **600-nm monochromatic light falls on a diffraction grating normally, and the first order bright band is observed at a angle** $\phi = \sin^{-1} 0.102$ **. If *a=b* , determine how many orders can be seen on the screen.**

Solution: $d\sin\theta = \pm m\lambda \quad (\text{m=0,1,2,3}\cdots\cdots)$

$$m = 1 \quad d\sin\theta = \lambda$$

$$m_{\max} = [\frac{d}{\lambda}] = [\frac{1}{0.102}] = 9$$

$a = b \quad \dfrac{d}{a} = 2,$ **Missing orders:** $\pm 2, \ \pm 4, \ \pm 6, \ \pm 8$

So, $0, \ \pm 1, \ \pm 3, \ \pm 5, \ \pm 7, \ \pm 9$ **orders can be seen on the screen**

Example:

λ = 0.5 μm 的单色光垂直入射到光栅上，测得第3级主极大的衍射角为30°，且第4级为缺级。求：(1)光栅常数 *d*；(2) 透光缝最小宽度*a*；(3) 对上述 *a*、*d* 屏幕上可能出现的谱线数目。

(1) $d\sin\theta_m = \pm m\lambda$ $\left\{\begin{array}{l} m=3 \\ \theta_3 = 30^{o} \end{array}\right\}$ $\boxed{d = 3\ \mu m}$

(2) 缺4级→ $\dfrac{d}{a} = 4$ $a_{min} = 0.75\ \mu m$

(3) $\sin\theta_m = m\dfrac{\lambda}{d} < 1$ $m_{max} < \left[\dfrac{d}{\lambda}\right] = 5$

$\boxed{m = 0,\ \pm 1,\ \pm 2,\ \pm 3,\ \pm 5}$

9条！

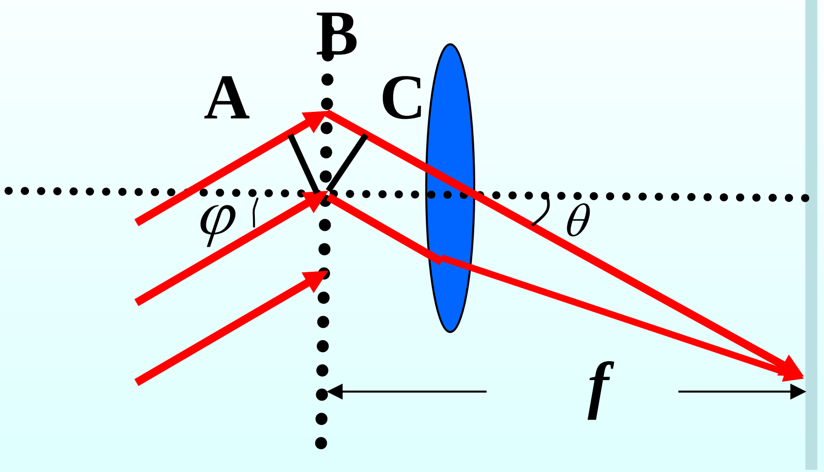
例 用每毫米有500条栅纹的衍射光栅观察钠光谱线 $\lambda = 590\text{nm}$ 。问：(1)光线垂直入射； (2)光线以30度角入射时，最多能看到第几级条纹？

解：(1) $d = \dfrac{1}{500} mm$ 最多条纹数由 $\sin\theta = 1$ 决定

$$d\sin\theta = m\lambda \qquad \therefore m = \left[\frac{d}{\lambda}\right] = [3.4] = 3$$

第三级条纹

(2) $$d(\sin\theta \pm \sin\varphi) = \pm m\lambda$$

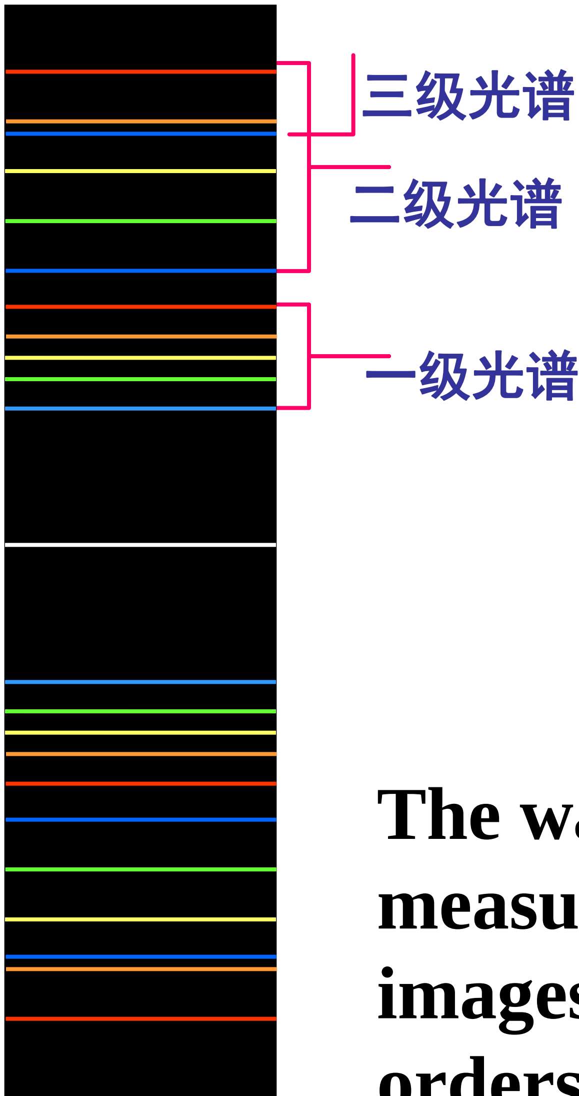


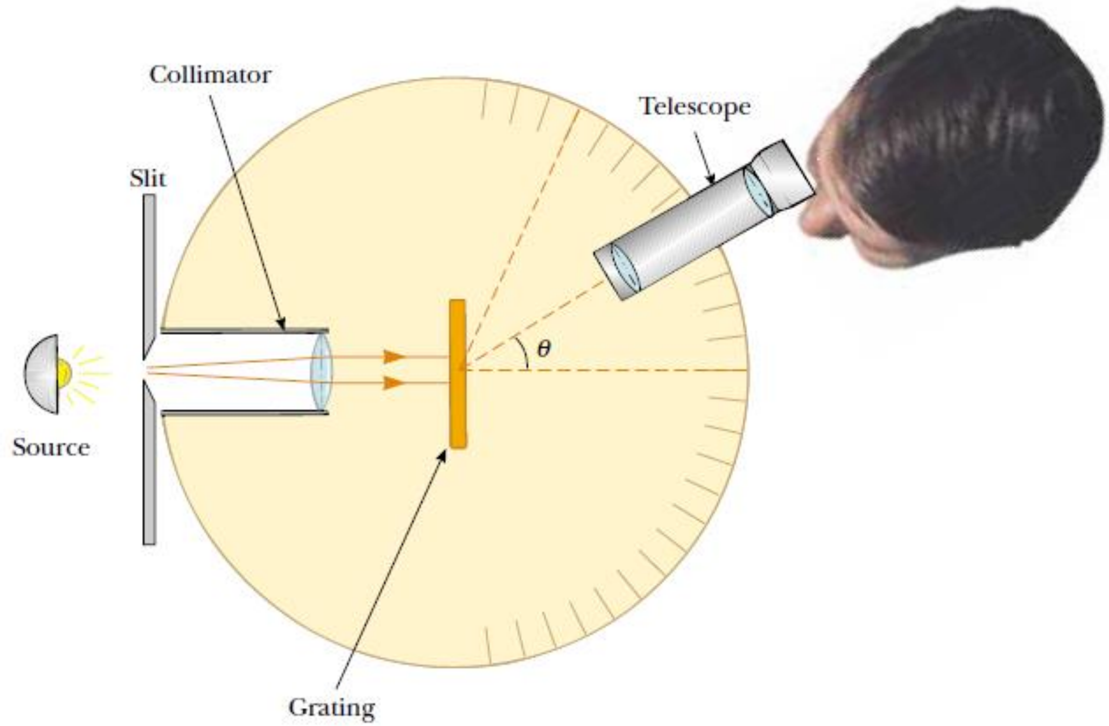
$$m_{\max} = \frac{d(-1-\sin\varphi)}{\lambda} = -5.08$$

第（负）五级条纹

4. Diffraction spectrum

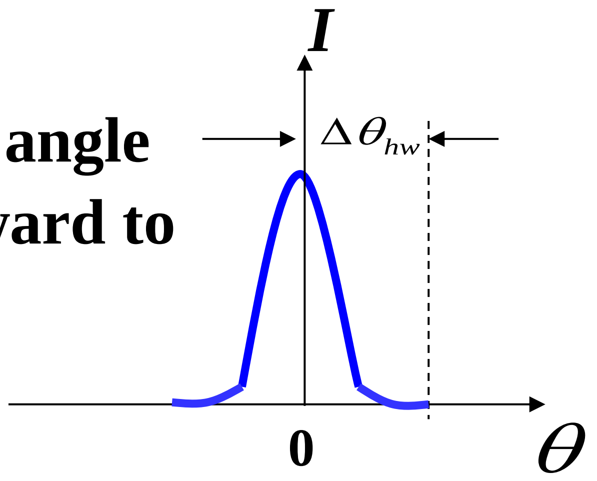
White bight strikes a grating, there will be a color spectrum





The wavelength can be determined by measuring the precise angles at which the images of the slit appear for the various orders.

*(a) Width of the Lines



Half-width of the central line as being the angle $\Delta\theta_{hw}$ from the center of the line at θ=0 outward to where the line effectively ends and darkness effectively begins with the first minimum.

$$Nd \sin \Delta\theta_{hw} = \lambda$$

Half-width of the central line $$\Delta\theta_{hw} \approx \sin \Delta\theta_{hw} = \frac{\lambda}{Nd}$$

Half-width of line at θ $$\Delta\theta_{hw} = \frac{\lambda}{Nd \cos \theta}$$

The widths of the lines decrease with an increase in the number N of rulings. Thus the grating with the larger value of N is able to distinguish between wavelengths.

*(b) Dispersion 色散

$$D = \frac{\Delta\theta}{\Delta\lambda} \qquad \text{(dispersion defined)}$$

$$D = \frac{m}{d \cdot \cos\theta} \qquad \text{(dispersion of a grating)}$$

To achieve a higher dispersion we must use a grating of smaller grating spacing *d* and work in a higher-order *m*.

*(c) Resolving power 光栅分辨本领

$$R = \frac{\lambda_{\text{avg}}}{\Delta\lambda} \qquad \text{(resolving power defined)}$$

$$R = Nm \qquad \text{(resolving power of a grating)}$$

To achieve high resolving power, we must use many rulings (*large N*).

*19.6 X-Ray diffraction

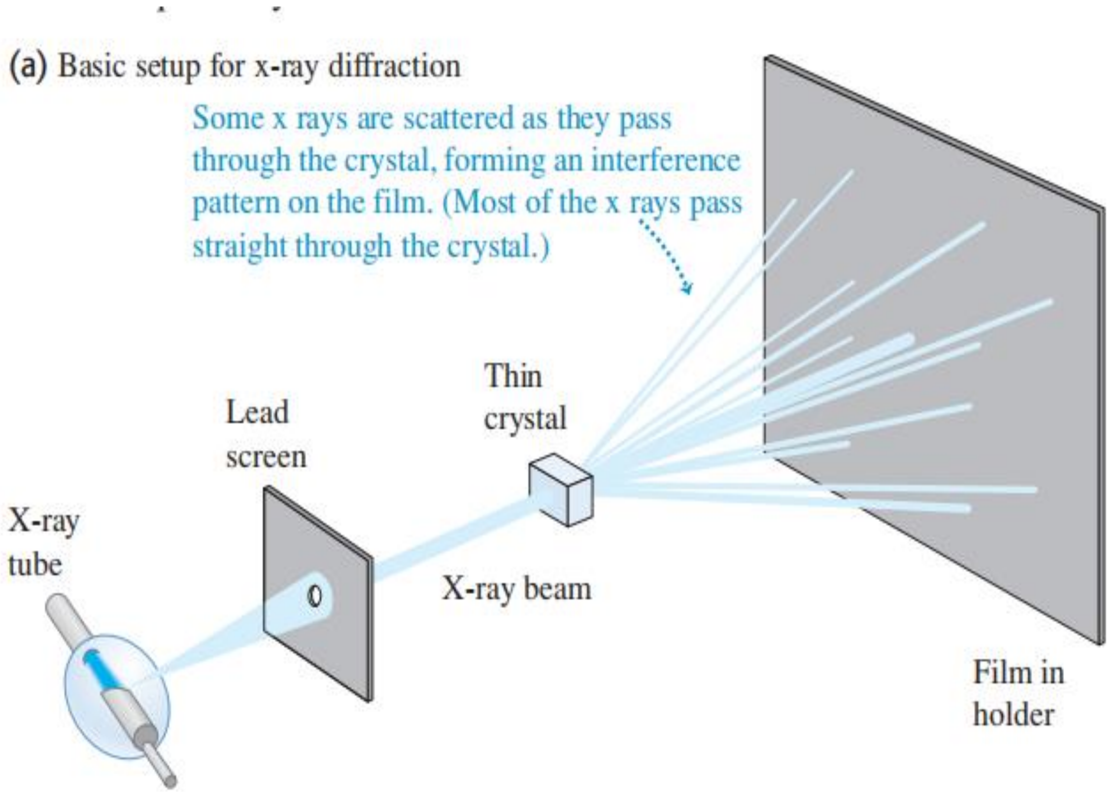
In 1885, Roentgen discovered a new type of radiation，called X-rays.

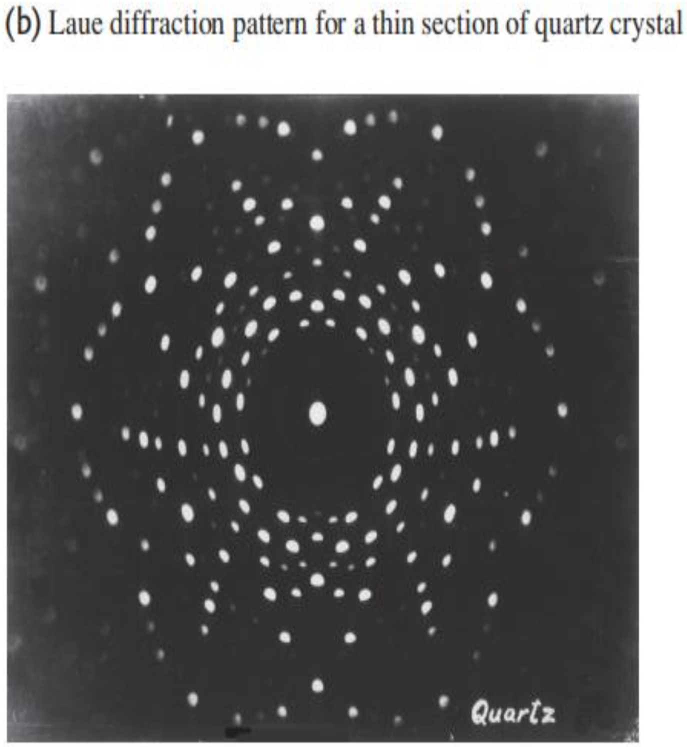
1901年12月10日，伦琴在瑞典的斯德哥尔摩第一个领取了诺贝尔物理奖。

X-rays are electromagnetic waves of very short wavelength (on the order of 0.1 nm). It would be impossible to construct a grating having such a small spacing by the cutting process.

1896年伦琴首次拍摄到他妻子手的X射线照片，其无名指上戴着一枚戒指.

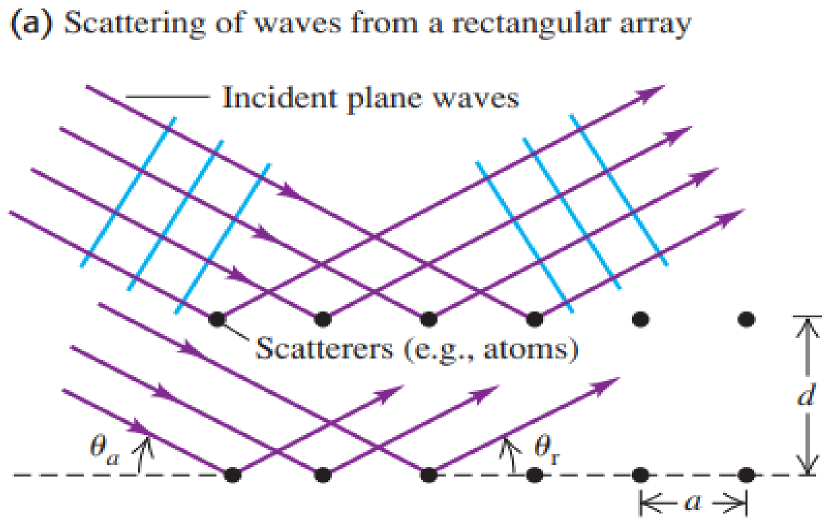
In 1913, Max von Laue (1879–1960) suggested that the regular array of atoms in a crystal could act as a three-dimensional diffraction grating for X-rays.

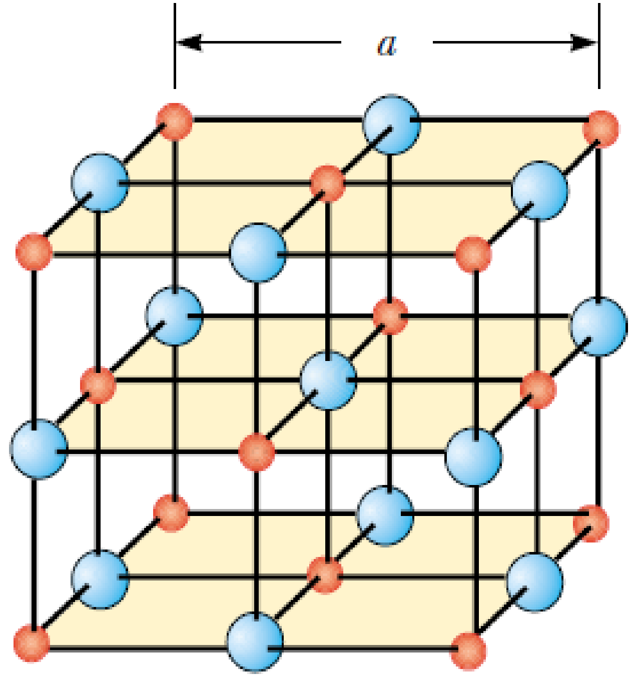




X-rays have a wave nature.

Bragg equation

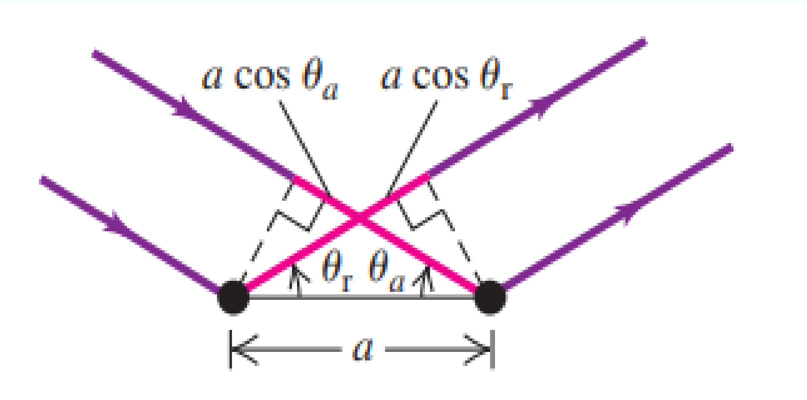




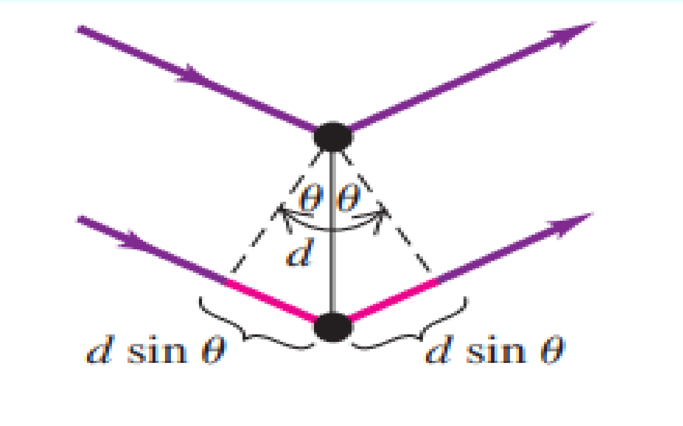
d：**lattice distance（晶面距离）**

θ **: glancing angle (掠射角)**

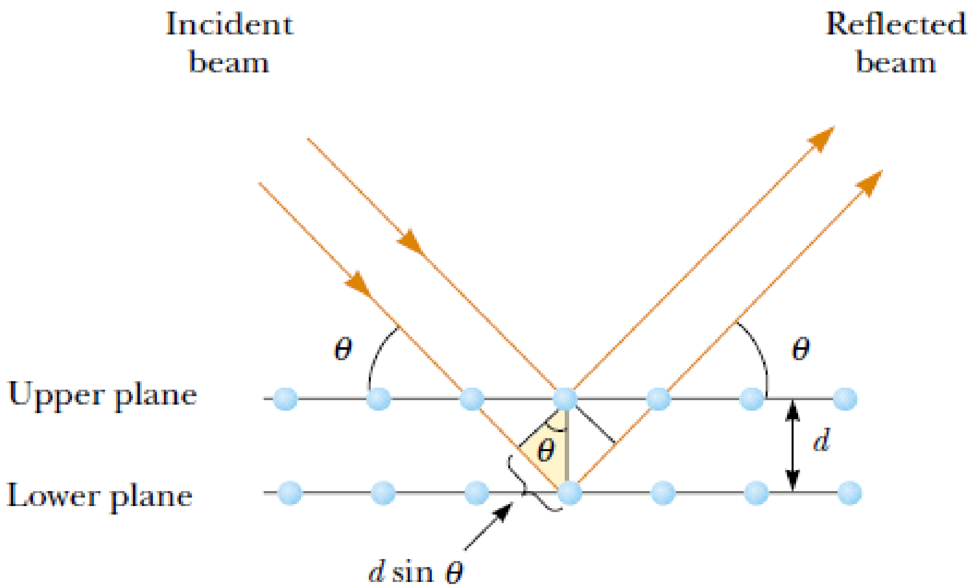
Scattering from adjacent atoms in a row



Scattering from atoms in adjacent rows



Bragg equation



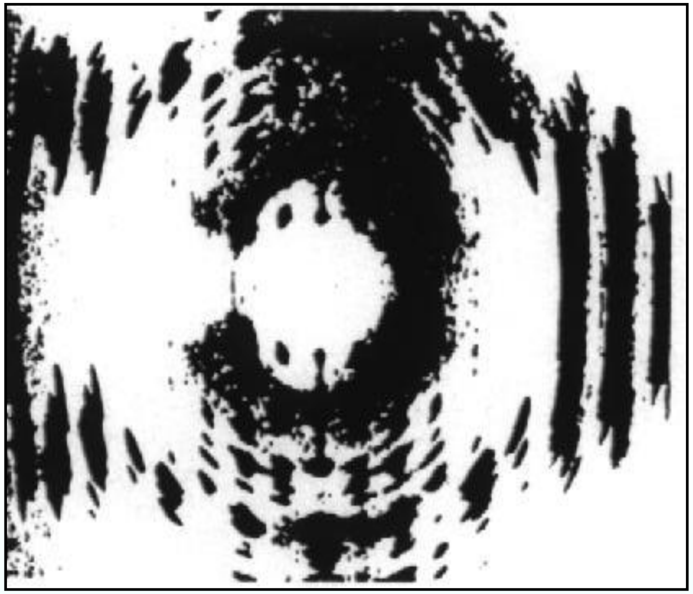
***d*：lattice distance（晶面距离）**

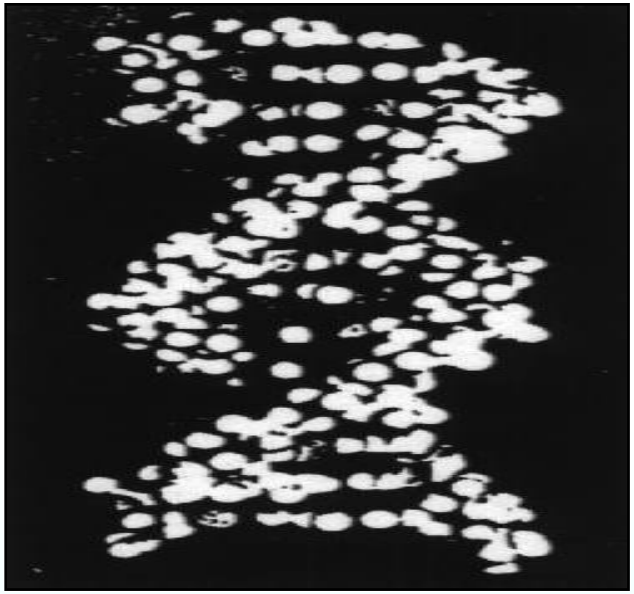
θ: glancing angle (掠射角)

Every atom can be thought of a wavelet of diffraction.

Two rays which are reflected from two subsequent planes of atoms constructively interfere:

$$2d\sin\theta = m\lambda \qquad m = 1, 2, 3\ldots$$ （**Bragg equation**）





DNA 晶体的X衍射照片

DNA 分子的双螺旋结构